Operational Amplifiers

1. Introduction

An operational amplifier is a **direct coupled amplifier** having **high voltage gain**.

It can be used to perform **mathematical operations on analog signals** — hence the name *operational* amplifier. The op-amp is available as **IC 741**, the general-purpose op-amp IC.

The IC 741 is a general-purpose operational amplifier available in an 8-pin Dual-In-Line Package (DIP). Pin 1 is identified by the notch (or a dot) on the top of the IC; pins are numbered anticlockwise when viewed from the top.

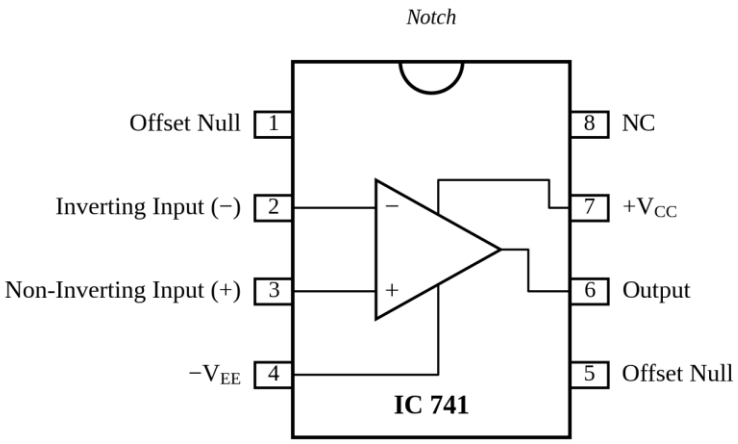


Figure 1: Pin diagram of IC 741 (8-pin DIP, top view)

Pin Description

Pin No.	Pin Name	Function
1	Offset Null	Used with pin 5 to remove output offset voltage
2	Inverting Input (−)	Signal at this pin appears inverted at output
3	Non-Inverting Input (+)	Signal at this pin appears in phase at output
4	$-V_{EE}$	Negative supply voltage (typically −15 V)
5	Offset Null	Used with pin 1 to remove output offset voltage
6	Output	Output terminal of the op amp

Pin No.	Pin Name	Function
7	$+V_{CC}$	Positive supply voltage (typically +15 V)
8	NC	No connection (not internally connected)

Offset Null Adjustment

A 10 kΩ potentiometer is connected between pins 1 and 5, with its wiper connected to $-V_{EE}$ (pin 4). With both inputs grounded, the potentiometer is adjusted until the output becomes zero.

Use of NC (Pin 8)

NC means **No Connection**. Pin 8 of IC 741 is not connected to any internal circuit, so it has no electrical function. It is provided for the following reasons:

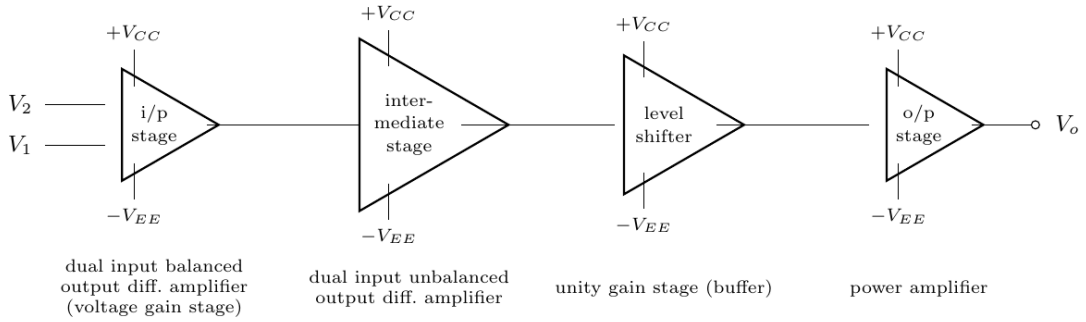
- **Standard package:** The 741 uses the standard 8-pin DIP package, but the op amp needs only 7 pins (2 inputs, 1 output, 2 supplies and 2 offset null). The eighth pin is left unused.
- **Pin compatibility:** Older op amps such as LM748 and LM301 are externally compensated and use pin 8 to connect a compensation capacitor. The 741 is internally compensated (on-chip 30 pF capacitor), so it does not need this pin. Keeping the same layout allows a 741 to replace these ICs in the same socket or PCB.
- **Mechanical support:** The pin is soldered to the PCB and helps to hold the IC firmly in place.

In practice: pin 8 is left unconnected. Some designers use it as a spare solder point for routing a PCB track, but no signal or supply should depend on it.

2. Internal Structure of IC 741[Unit 2 Topic]

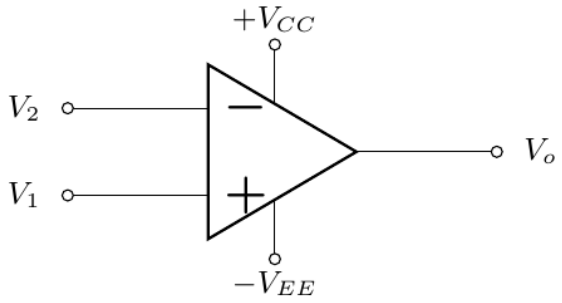
IC 741 internally consists of **four stages**:

i. **Input stage** — dual input balanced output stage
ii. **Intermediate stage** — dual input unbalanced output stage
iii. **Level shifter**
iv. **Output stage**



- The **input stage** is a dual input balanced output differential amplifier; the **intermediate stage** is a dual input unbalanced output differential amplifier.
- Two differential amplifiers are used in the internal circuit of IC 741 to achieve **high voltage gain** and a **high CMRR**.
- The unbalanced output of the intermediate stage has a **non-zero DC biasing voltage**, which can disturb the operating point of the transistors in the output stage and hence cause distortion in the final output. The **level shifter** is used to eliminate the DC biasing voltage present at the output of the intermediate stage.
- The level shifter removes the DC *only* due to the biasing supply, not the input DC. Input DC, if applied, should reach the output and should not be suppressed by the level shifter.
- The level shifter has **high input impedance**, which prevents the loading effect on the intermediate stage.
- The **output stage** is a complementary symmetry push-pull power amplifier. A power amplifier is used so that the op-amp IC is able to supply **large output power** to the load.

3. Op-Amp Symbol



For an op-amp, the net input is $(V_1 - V_2)$, and $\frac{V_o}{V_1 - V_2}$ is the **open loop voltage gain**:

$$A_{OL} = \frac{V_o}{V_1 - V_2} = \frac{-V_o}{V_2 - V_1} \qquad \Rightarrow \qquad V_o = A_{OL}(V_1 - V_2)$$

A_{OL} is a large positive value.

- If $V_2 = 0$, then $V_o = A_{OL}V_1$, and hence V_o and V_1 are **in phase**. Therefore V_1 is the **non-inverting input**.
- If $V_1 = 0$, then $V_o = -A_{OL}V_2$, and hence V_o and V_2 are 180° **out of phase**. Therefore V_2 is the **inverting input**.

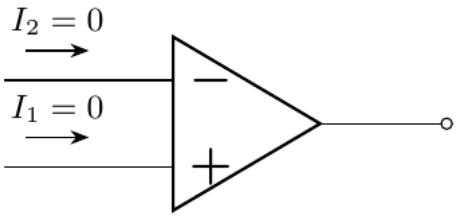
4. Properties of an Ideal Op-Amp

(i) Open loop voltage gain is infinite

$$A_{OL} = \infty$$

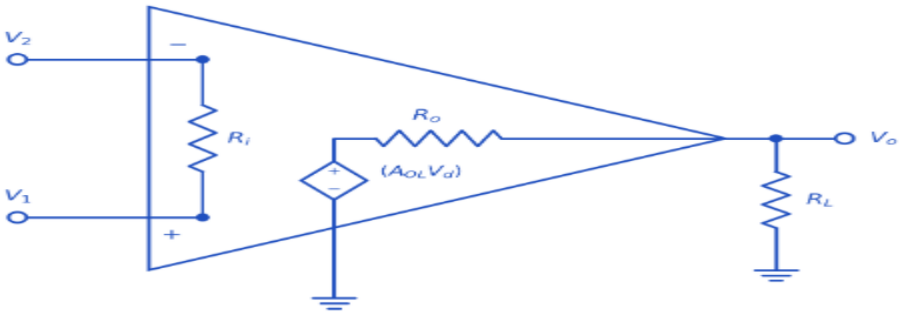
(ii) Input resistance is infinite

$R_i = \infty$, hence the input currents of an ideal op-amp are considered to be **zero**.



(iii) Output resistance is zero

$R_o = 0$, hence the output voltage of an ideal op-amp is **independent of the load** R_L.

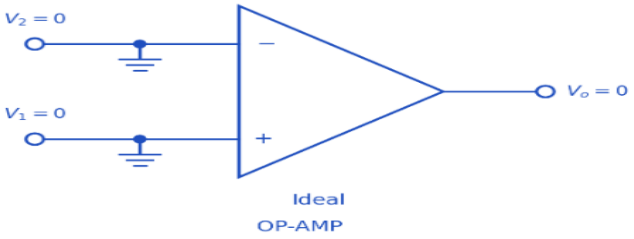


According to the values of R_i and R_o, an ideal op-amp is an **ideal voltage amplifier**, or a **voltage dependent voltage source**.

(iv) Bandwidth is infinite

$BW = \infty$, hence an ideal op-amp can amplify a signal of **any frequency**.

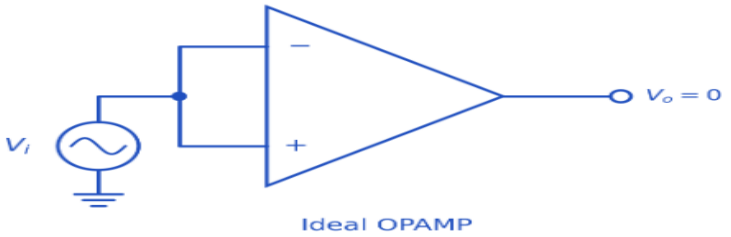
(v) Zero offset



If $V_1 = 0$ and $V_2 = 0$, then $V_o = 0$.

(vi) CMRR is infinite

$CMRR = \infty$, hence an ideal op-amp will **fully reject the common mode input**: if $V_1 = V_2$ then $V_o = 0$.



(vii) Slew rate is infinite

$SR = \infty$.

> An ideal op-amp is an **ideal difference amplifier** — it amplifies only the difference between the two inputs.

5. Properties of a Practical Op-Amp (IC 741)

i. $A_{OL} = 10^6$
ii. $R_i = 1\,\text{M}\Omega$ to $2\,\text{M}\Omega$
iii. $R_o = 50\,\Omega$ to $100\,\Omega$
iv. Open loop $BW = 5$ Hz (bandwidth without feedback)
v. Non-zero offset — if $V_1 = 0$ and $V_2 = 0$ then $V_o \neq 0$
vi. $CMRR = 10^6$ or 120 dB
vii. Slew rate, $SR = 0.5$ to $1\ \text{V}/\mu\text{s}$

> IC 741 has a **smaller open loop bandwidth**, but the bandwidth can be increased with the help of **negative feedback**.

CMRR in decibels

$$(CMRR)_{\text{dB}} = 20\log\left(\frac{A_{DM}}{A_{CM}}\right) \qquad (CMRR)_{\text{dB}} = (A_{DM})_{\text{dB}} - (A_{CM})_{\text{dB}}$$

6. Slew Rate

The **slew rate** is the maximum rate at which the output voltage of an op-amp can change. Unit: **volt/μs**. For example, $SR = 1\ \text{V}/\mu\text{s}$.

The output voltage of an op-amp cannot change at a rate greater than the slew rate. If the input signal to the op-amp has faster changes, then the output voltage cannot follow the

changes in the input and hence the output signal gets distorted. This is called **slew rate induced distortion**.

Condition to prevent SR induced distortion

Consider a closed loop op-amp circuit having gain A_{CL}, and let $V_i = V_m \sin\omega_0 t$. Then:

$$V_o = A_{CL} \times V_i = A_{CL} V_m \sin\omega_0 t$$

$$\frac{dV_o}{dt} = A_{CL} V_m \omega_0 \cos\omega_0 t \qquad \Rightarrow \qquad \left.\frac{dV_o}{dt}\right|_{\max} = A_{CL} V_m \omega_0 \quad \text{(volt/sec)}$$

$$\left.\frac{dV_o}{dt}\right|_{\max} = \frac{A_{CL} V_m \omega_0}{10^6} \quad \text{(volt/}\mu\text{s)} \qquad (1)$$

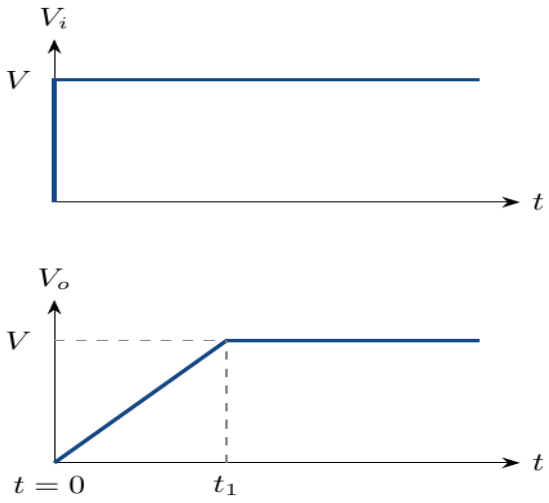
To prevent distortion, $\left.\frac{dV_o}{dt}\right|_{\max} \leq SR$, so:

$$\frac{A_{CL} V_m \omega_0}{10^6} \leq SR$$

If A_{CL} is not provided, then it is considered to be **unity**.

Practical measurement of slew rate

Practically the slew rate is measured by applying a large signal **step input** to a **unity gain** op-amp circuit.

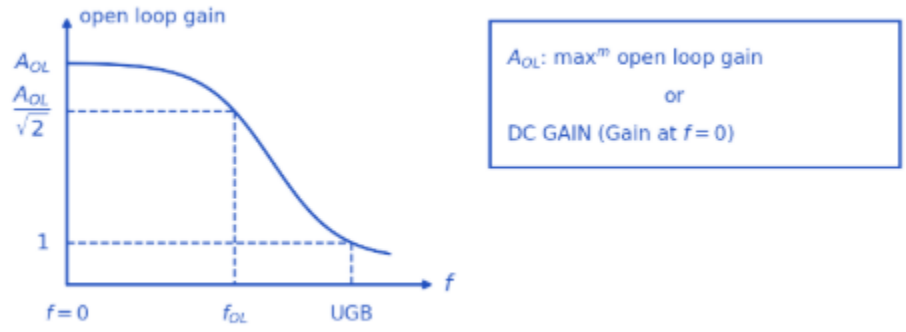


output obtained using a unity gain amplifier (voltage follower)

$$\text{Slew rate} = \text{slope} = \frac{\Delta V}{\Delta t} = \frac{V - 0}{t_1 - 0} = \frac{V}{t_1}$$

7. Frequency Response of IC 741

The frequency response is a plot of **open loop voltage gain versus frequency**.



A_{OL} is the **maximum open loop gain**, or the **DC gain** (gain at $f = 0$). An op-amp is a direct coupled amplifier, hence the voltage gain remains constant at low frequencies (no large capacitor is present). But the voltage gain **decreases at high frequencies** due to a **compensation capacitor**.

The **compensation capacitor** (pF) is used in the internal circuit of IC 741 to provide **dominant pole compensation**, which increases the stability of the op-amp. IC 741 is called an **internally compensated op-amp IC** because the compensation capacitor is inbuilt. IC 708 is a **special purpose op-amp IC** — the compensation capacitor has to be connected externally.

The corner frequency f_{OL}

f_{OL} is called the **cutoff frequency**, **3 dB frequency**, **corner frequency**, **break frequency**, **open loop bandwidth** or **dominant pole frequency**.

For IC 741: $A_{OL} = 10^6$ and $f_{OL} = 5$ Hz (dominant pole frequency).

The voltage gain of the op-amp at high frequencies is calculated from:

$$|A_V| = \frac{A_{OL}}{\sqrt{1 + (f/f_{OL})^2}} \qquad A_V = \frac{A_{OL}}{1 + j\, f/f_{OL}}$$

8. Unity Gain Bandwidth (UGB)

The **unity gain bandwidth** is the high frequency at which the open loop gain of the op-amp becomes **unity**. At $f = \text{UGB}$, $|A_V| = 1$:

$$1 = \frac{A_{OL}}{\sqrt{1 + \left(\frac{\text{UGB}}{f_{OL}}\right)^2}}$$

Since UGB $\ggg f_{OL}$, we have $\frac{\text{UGB}}{f_{OL}} \ggg 1$, so:

$$1 = \frac{A_{OL}}{\frac{\text{UGB}}{f_{OL}}} = \frac{A_{OL} \times f_{OL}}{\text{UGB}} \qquad \Rightarrow \qquad \text{UGB} = A_{OL} \times f_{OL}$$

This is also called the **open loop gain-bandwidth product** of the op-amp.

> UGB represents the **highest frequency** up to which the op-amp can provide amplification. For IC 741: UGB $= 10^6 \times 5 = 5$ MHz.

Worked problem

Q. An op-amp has a DC gain of 10^4 and a gain of 1000 at a frequency of 500 Hz. Calculate the corner frequency and the UGB of the op-amp.

$$1000 = \frac{10^4}{\sqrt{1 + (500/f_{OL})^2}} \qquad \Rightarrow \qquad 1 + \left(\frac{500}{f_{OL}}\right)^2 = 10^2$$

$$\frac{(500)^2}{(f_{OL})^2} = 100 \qquad \Rightarrow \qquad (f_{OL})^2 = \frac{500 \times 500}{100} \qquad \Rightarrow \qquad f_{OL} \approx 50 \text{ Hz}$$

$$\text{UGB} = A_{OL} \times f_{OL} = 10^4 \times 50 = 500 \text{ kHz} = 0.5 \text{ MHz}$$

Effects of the compensation capacitor

i. It **improves the stability** of the op-amp.
ii. It **reduces the gain** of the op-amp at high frequencies.
iii. It **reduces the slew rate** of the op-amp (a capacitor does not allow sudden changes in voltage).

9. Transfer Characteristics of an Op-Amp

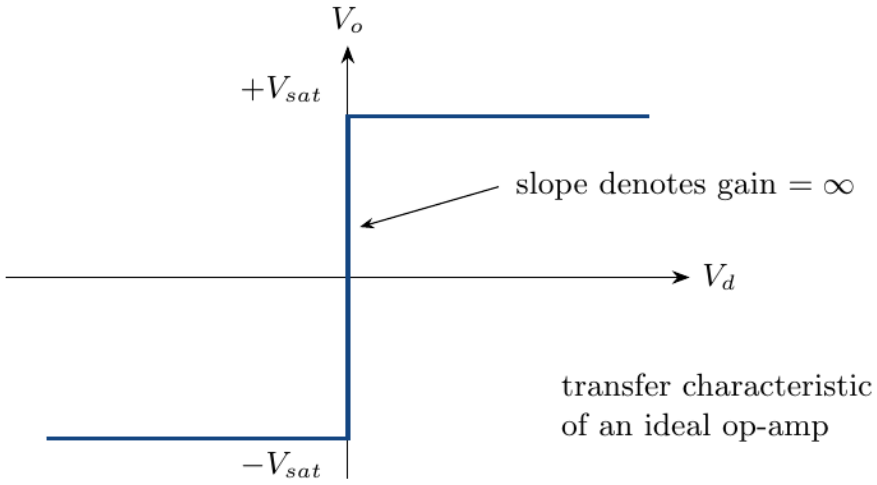
A plot of V_o versus the difference input V_d, where $V_d = V_1 - V_2$:

$$V_o = A_{OL} V_d = A_{OL}(V_1 - V_2)$$

(a) Ideal op-amp

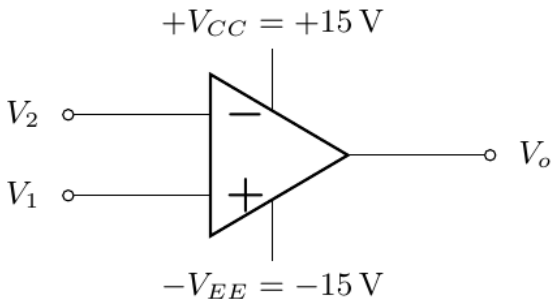
For an ideal op-amp $A_{OL} = \infty$:

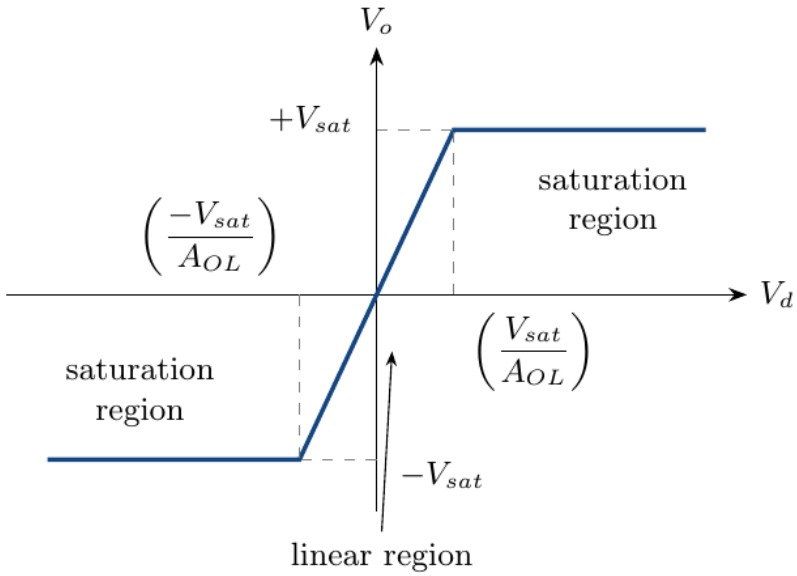
- If V_d is positive, $V_o = +\infty \equiv +V_{CC} = +V_{sat}$ (most positive output).
- If V_d is negative, $V_o = -\infty \equiv -V_{EE} = -V_{sat}$ (most negative output).



(b) Practical op-amp

For a practical op-amp $A_{OL} = 10^6$, so $V_o = 10^6 V_d$.





If the difference input is very small, i.e. if it lies in the range $\left(\frac{-V_{sat}}{A_{OL}}\right)$ to $\left(\frac{+V_{sat}}{A_{OL}}\right)$, then the output voltage varies **linearly** with the difference input. If the difference input is outside the above range, then the output voltage becomes $+V_{sat}$ or $-V_{sat}$, i.e. the output gets **saturated**.

10. Operating Regions of an Op-Amp

An op-amp can operate in (i) the **linear region** or (ii) the **saturation region**.

- When the op-amp is used **without feedback** or with **positive feedback**, the difference input V_d will be large, and hence the op-amp can be considered in the **saturation region**. Then $V_o = \pm V_{sat}$.
- When the op-amp is used with **negative feedback**, the difference input V_d becomes very small, because the feedback signal opposes the applied input. Hence it can be considered in the **linear region**.

11. Virtual Short Circuit and Virtual Ground

Virtual short circuit condition

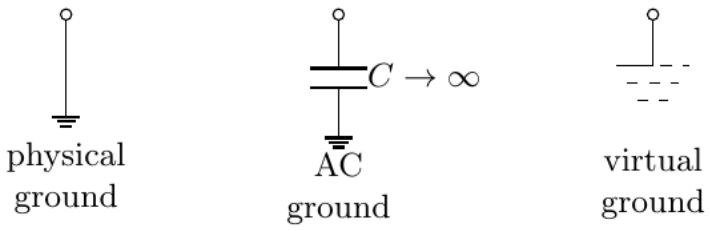
When the op-amp is in the linear region, the difference input V_d will be very small (μV range). Hence in mathematical analysis this difference input can be approximated to zero:

$$V_d \cong 0 \quad \Rightarrow \quad V_1 - V_2 \cong 0 \quad \Rightarrow \quad V_1 \cong V_2$$

When the op-amp is in the linear region, the two input terminals of the op-amp will be at approximately equal voltage **without any physical short circuit** between them. Hence the two input terminals are said to be **virtually shorted**.

Virtual ground

If a virtual short circuit is present between two nodes A and B, and if node B is **physically grounded**, then the voltage at node A also becomes zero — node A gets **virtually grounded**. If a node voltage becomes zero *without* physically connecting it to ground, then the node is said to be at **virtual ground**.

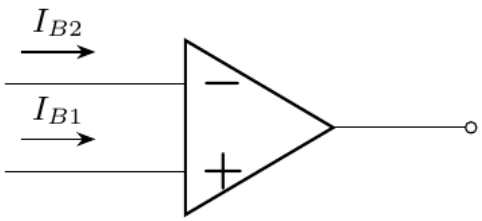


Virtual short circuit and **virtual ground** are used in the analysis of **negative feedback op-amp circuits**, in which the op-amp is assumed to be in the **linear region**.

12. Important Definitions (IC 741)

(i) Input bias current I_B

It is the **average** of the two input currents of the op-amp.



$$I_B = \frac{I_{B1} + I_{B2}}{2}$$ — ranges in hundreds of nA

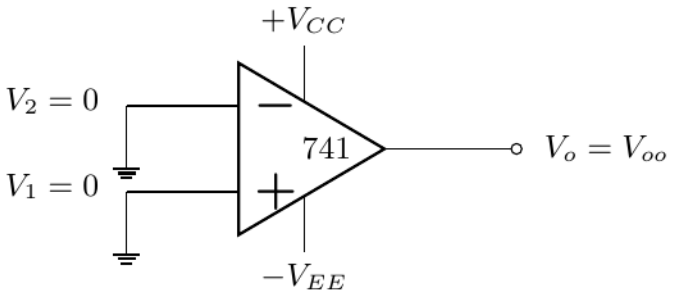
(ii) Input offset current I_{io}

It is the **difference** of the two input currents of the op-amp.

$$I_{io} = |I_{B1} - I_{B2}|$$ — ranges in tens of nA

(iii) Output offset voltage V_{oo}

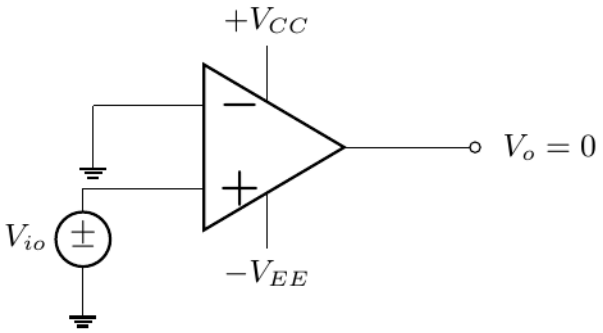
It is the output voltage of the op-amp when **both inputs are zero**.



V_{oo} ranges in **volts**. In the worst case $V_{oo} = \pm V_{sat}$, and in such a case the op-amp is of no use. V_{oo} is due to dissimilarities in the transistors and mismatch in the resistor values in the internal circuit of the op-amp.

(iv) Input offset voltage V_{io}

It is a **non-zero voltage** which should be applied as the difference input to make the output voltage of the op-amp **zero**.



V_{io} ranges in μ**V**. V_{oo} and V_{io} can vary due to a change in supply voltages and a change in temperature.

(v) Power supply rejection ratio (PSRR)

It is a measure of the change in input offset voltage due to a change in supply voltage.

$$PSRR = \frac{\Delta V_{io}}{\Delta V_s} \qquad (PSRR)_{\text{dB}} = 20\log\left(\frac{\Delta V_s}{\Delta V_{io}}\right)$$

where ΔV_{io} is the change in input offset voltage and ΔV_s the change in supply voltage.

In the decibel expression the orientation is inverted only to obtain a positive value.

Worked problem

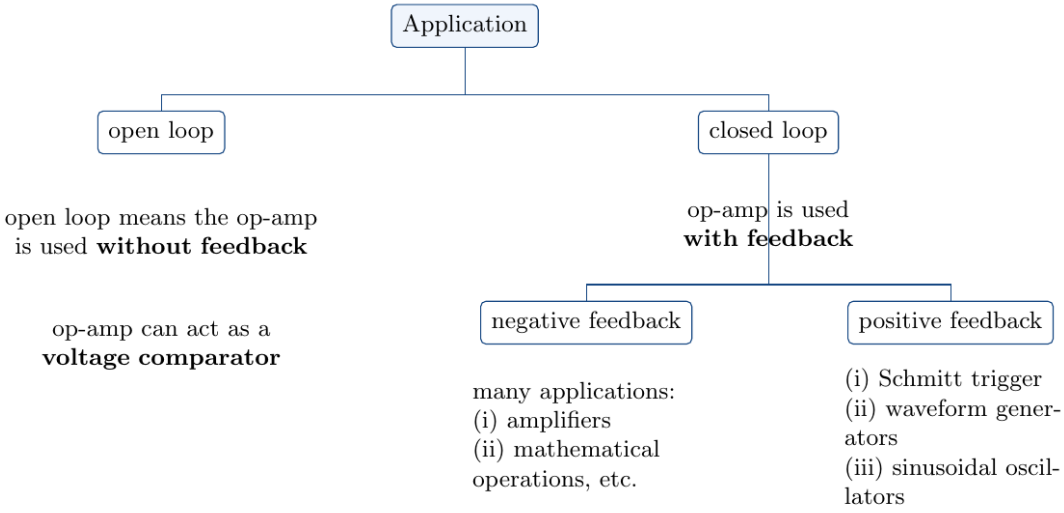
Q. An op-amp has a PSRR of 120 dB. If the supply voltage varies by ± 1 V, calculate the change in input offset voltage.

$$120 = 20\log\left(\frac{\pm 1}{\Delta V_{io}}\right) \quad \Rightarrow \quad 6 = \log\left(\frac{\pm 1}{\Delta V_{io}}\right) \quad \Rightarrow \quad \frac{\pm 1}{\Delta V_{io}} = 10^6$$

$$\Delta V_{io} = \pm 1\ \mu\text{V}$$

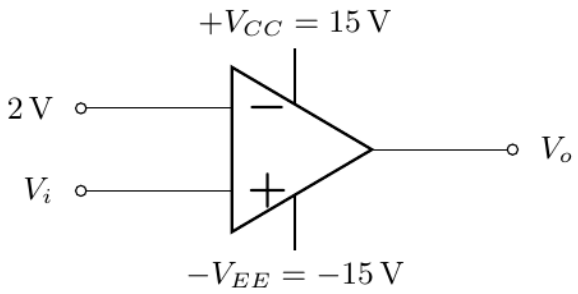
Applications:

1. Classification of Op-Amp Applications



2. Op-Amp as a Comparator — Worked Problem

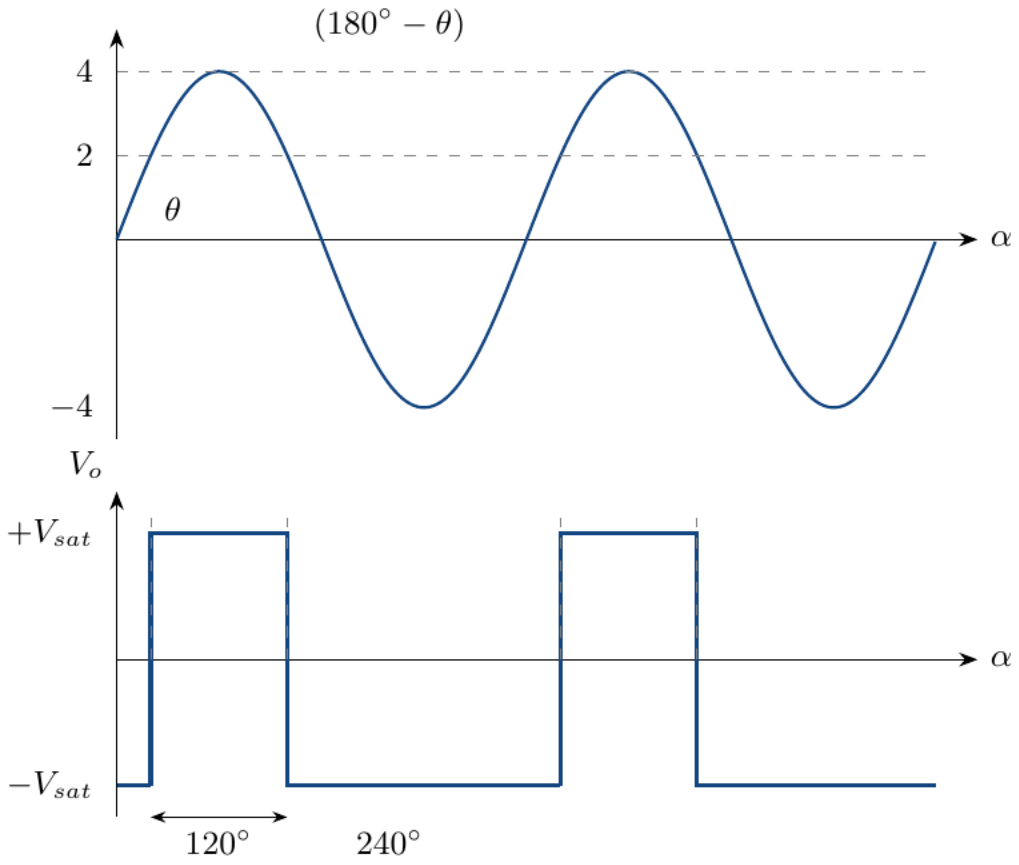
Q. For the op-amp shown below, the input is sinusoidal with a peak value of 4 V. Calculate (a) the duty cycle of the output waveform and (b) the average value of the output, if the supply voltages are ± 15 V.



Solution. When the op-amp is used without feedback it operates in the saturation region and acts as a **comparator**. Here V_i is compared with 2 V, so $V_d = (V_i - 2)$:

$$\text{if } V_i > 2 \;\Rightarrow\; V_d = +\text{ve} \;\Rightarrow\; V_o = +V_{sat}$$

$$\text{if } V_i < 2 \;\Rightarrow\; V_d = -\text{ve} \;\Rightarrow\; V_o = -V_{sat}$$



The output changes state when $V_i = 2$ V, i.e. when $4\sin\theta = 2$:

$$\theta = \sin^{-1}\left(\frac{2}{4}\right) = 30°$$

The output is $+V_{sat}$ from θ to $(180° - \theta)$, i.e. for a duration of $(180° - 2\theta) = 120°$.

(a) Duty cycle

$$\text{Duty cycle of } V_o = \frac{\text{duration when } V_o \text{ is } +V_{sat}}{\text{time period}} = \frac{120°}{360°} = \frac{1}{3}$$

(b) Average value

$$\text{Average } V_o = \frac{\text{area}}{\text{time period}} = \frac{120 \times (+V_{sat}) + 240 \times (-V_{sat})}{360} = \frac{120 \times 15 - 240 \times 15}{360}$$

$$= -5 \text{ V}$$

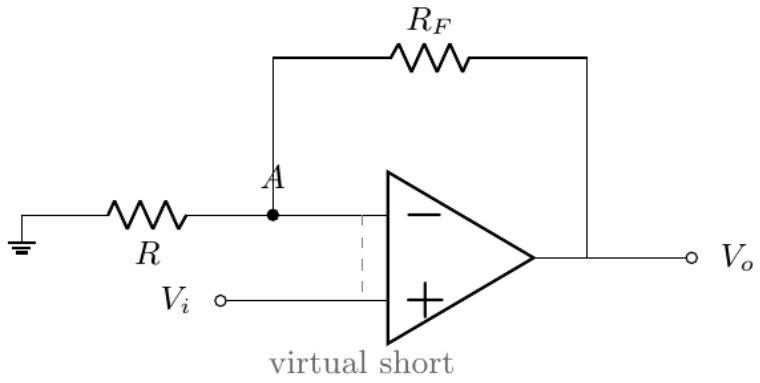
3. Negative Feedback Applications of the Op-Amp

The following assumptions are made in the analysis of negative feedback op-amp circuits:

- The op-amp is assumed to be in the **linear region**, and hence the two input terminals are **virtually shorted**.
- The **input currents** of the op-amp are **neglected**.

4. Non-Inverting Amplifier

A circuit in which V_o and V_i are **in phase** is a non-inverting amplifier.



R and R_F are in series and hence form a **voltage divider**. This voltage divider results in **negative feedback**.

Using the voltage division principle:

$$V_A = \frac{V_o \times R}{R + R_F}$$

The difference input to the op-amp is $V_d = V_i - V_A$.

Test of negative feedback

$$\text{If } V_o \uparrow \rightarrow\ V_A \uparrow \rightarrow\ V_d \downarrow \rightarrow\ V_o \downarrow$$

As an increase in output causes a decrease in output, the circuit is said to have **negative feedback**. (If an increase in output causes a further increase in output, then it is called **positive feedback**.)

Closed loop gain

Due to negative feedback the difference input V_d becomes very small:

$$V_d \cong 0 \qquad \Rightarrow \qquad V_i - V_A \cong 0 \qquad \Rightarrow \qquad V_A \cong V_i$$

Hence the concept of virtual short can be applied:

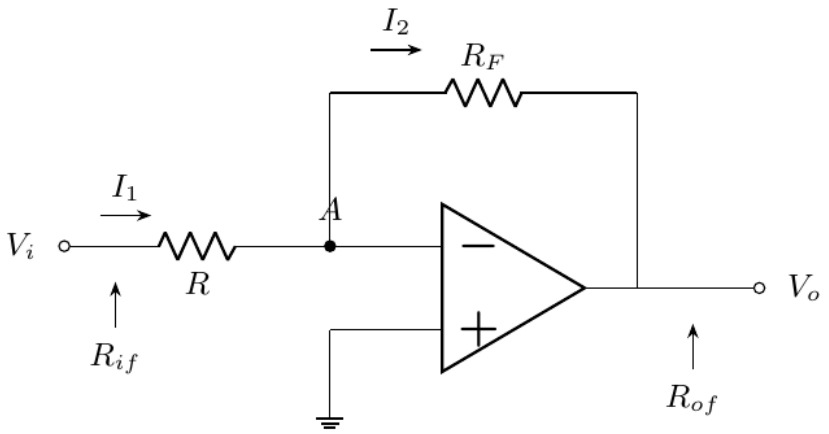
$$\frac{V_o R}{R + R_F} = V_i \qquad \Rightarrow \qquad V_o = \left(\frac{R + R_F}{R}\right) V_i$$

$$V_o = \left(1 + \frac{R_F}{R}\right) V_i \qquad \Rightarrow \qquad A_{CL} = \frac{V_o}{V_i} = \left(1 + \frac{R_F}{R}\right)$$

V_o and V_i are **in phase**.

5. Inverting Amplifier

A circuit in which the output and input are 180° **out of phase** is an inverting amplifier.



R and R_F result in **negative feedback**. Due to the virtual short circuit, the voltage at node A also becomes zero, i.e. node A gets **virtually grounded**: $V_A = 0$.

KCL at node A:

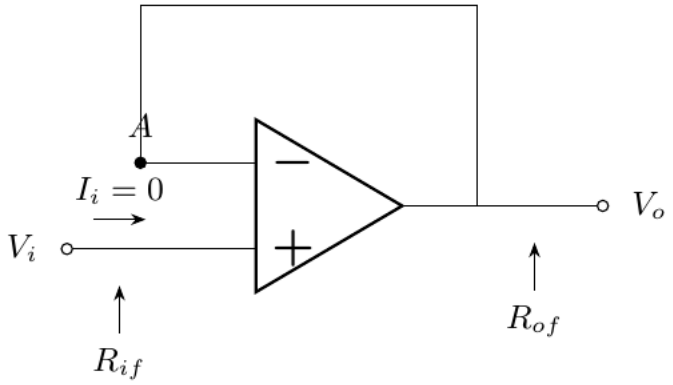
$$I_1 + I_2 = 0 \qquad \Rightarrow \qquad \left(\frac{V_i - V_A}{R}\right) + \left(\frac{V_o - V_A}{R_F}\right) = 0 \qquad \Rightarrow \qquad \frac{V_i}{R} + \frac{V_o}{R_F} = 0$$

$$V_o = \frac{-R_F}{R} V_i \qquad \Rightarrow \qquad A_{CL} = \frac{V_o}{V_i} = \frac{-R_F}{R}$$

V_o and V_i are 180° out of phase.

If $R_F = R$, then $V_o = -V_i$ — such a circuit is a **unity gain inverting amplifier**.

6. Voltage Follower



Due to the **physical short circuit**: $V_A = V_o$. Due to the **virtual short circuit**: $V_A = V_i$. Hence $V_o = V_i$ — the output voltage follows the input voltage, and $\frac{V_o}{V_i} = A_{CL} = 1$.

A voltage follower is also called a **unity gain non-inverting amplifier**.

Large signal bandwidth

The large signal bandwidth is calculated from the slew rate condition:

$$\frac{A_{CL} \times V_m \times \omega_0}{10^6} \leq SR \qquad \text{with } A_{CL} = 1 \qquad \Rightarrow \qquad \frac{V_m \omega_0}{10^6} \leq SR$$

$$f_0 = \frac{SR \times 10^6}{2\pi V_m}$$

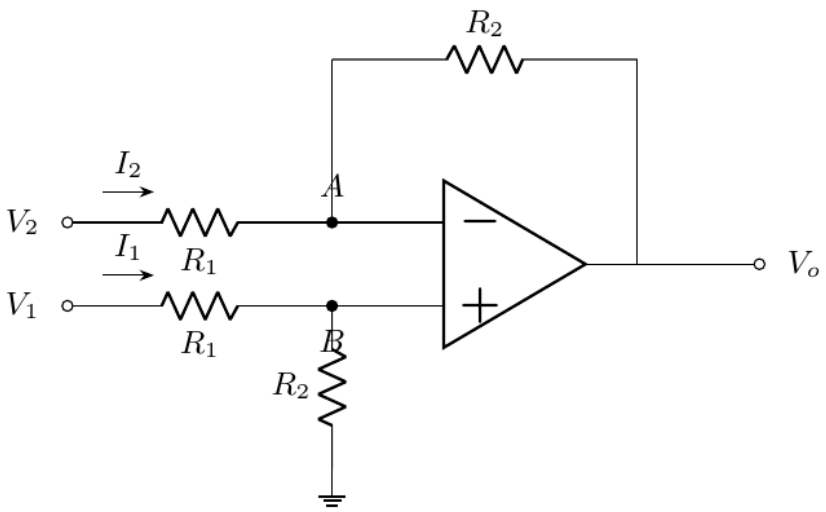
(The same approach applies to inverting and non-inverting amplifiers.)

In op-amp circuits: **large signal bandwidth** depends upon the **slew rate**; **small signal bandwidth** depends upon the **UGB**.

Applications of the voltage follower

a. Voltage buffer for **impedance matching**
b. Used in **sample and hold** circuits
c. Used in the **instrumentation amplifier** to prevent loading of the transducer bridge
d. Used in **first order Butterworth filters**

7. Difference Amplifier



Due to the virtual short, $V_A = V_B$:

$$\frac{V_2R_2 + V_oR_1}{R_1 + R_2} = \frac{V_1R_2}{R_1 + R_2} \quad \Rightarrow \quad V_2R_2 + V_oR_1 = V_1R_2$$

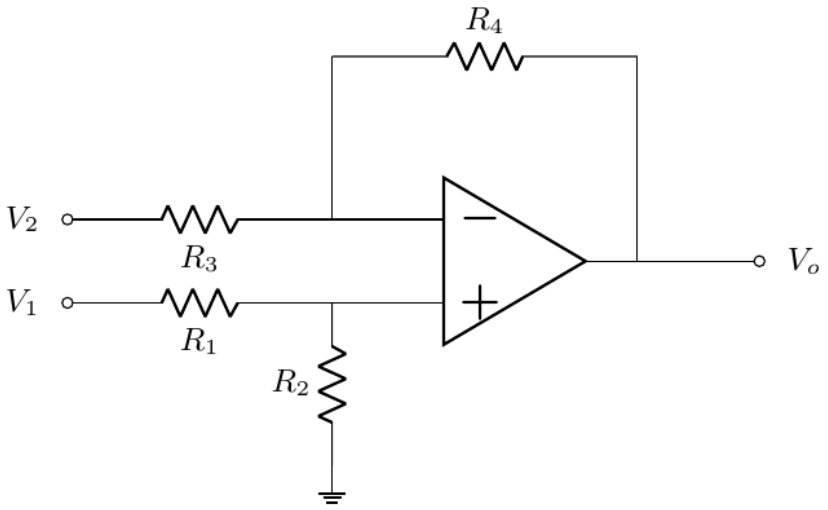
$$V_oR_1 = -V_2R_2 + V_1R_2 \quad \Rightarrow \quad V_o = \frac{R_2}{R_1}(V_1 - V_2)$$

$$\frac{V_o}{V_1 - V_2} = \frac{R_2}{R_1}$$

The circuit amplifies the difference $(V_1 - V_2)$, hence it is a **difference amplifier**.

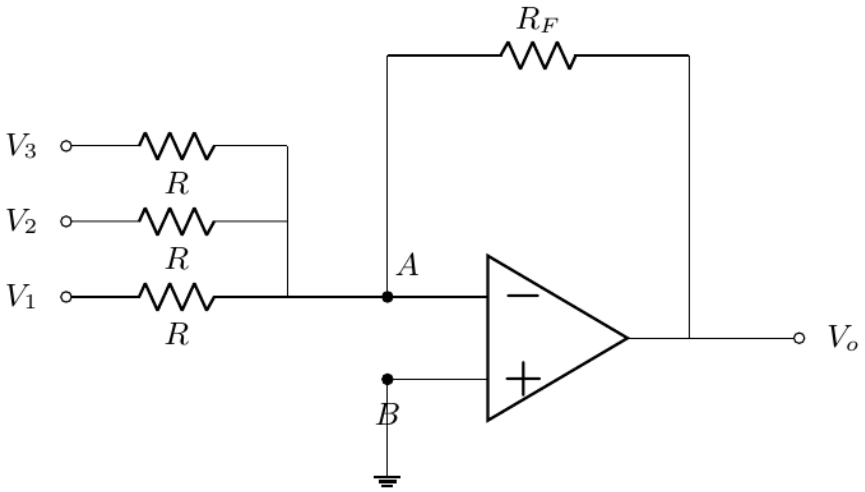
An alternative difference amplifier

The following circuit will also act as a difference amplifier if $\frac{R_2}{R_1} = \frac{R_4}{R_3}$



A difference amplifier can act as a **subtractor** if $R_1 = R_2$: $V_o = V_1 - V_2$.

8. Inverting Summing Amplifier



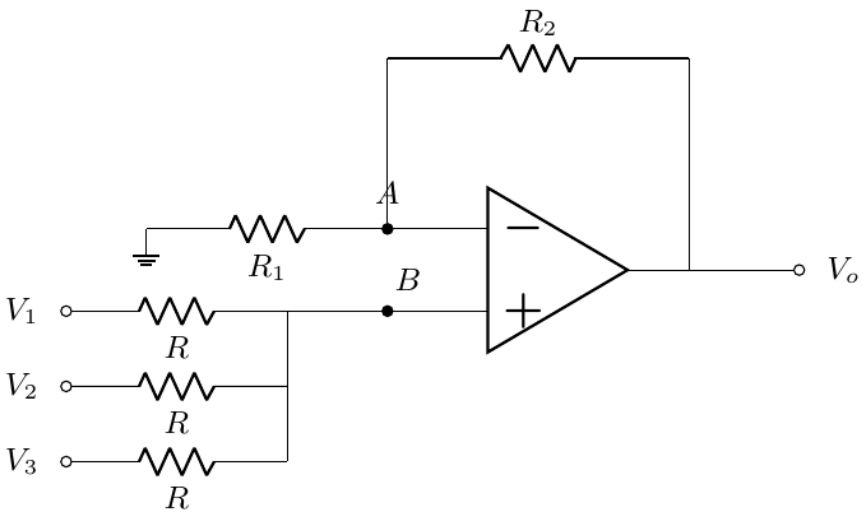
Due to the virtual ground, $V_A = 0$. KCL at node A:

$$\frac{V_1 - 0}{R} + \frac{V_2 - 0}{R} + \frac{V_3 - 0}{R} + \frac{V_o - 0}{R_F} = 0$$

$$V_o = -\frac{R_F}{R}(V_1 + V_2 + V_3)$$

The three inputs get added and their sum is amplified by a factor (R_F/R), and a phase shift of 180° also occurs. If $R_F = R$, then $V_o = -(V_1 + V_2 + V_3)$ — such a circuit is called an **inverting type adder**.

9. Non-Inverting Summing Amplifier



KCL at node B:

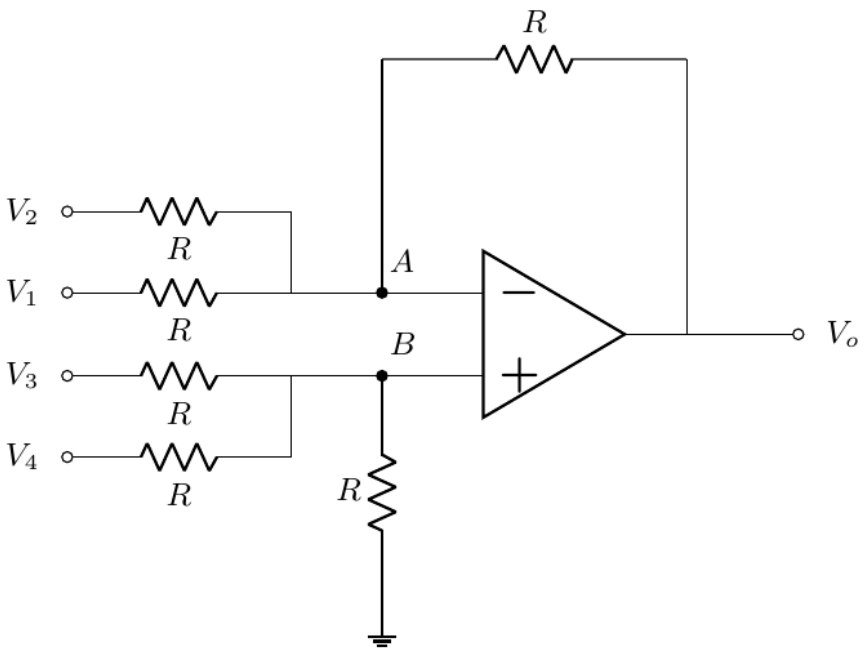
$$\frac{V_1 - V_B}{R} + \frac{V_2 - V_B}{R} + \frac{V_3 - V_B}{R} = 0 \qquad \Rightarrow \qquad V_B = \frac{V_1 + V_2 + V_3}{3}$$

Using the voltage division rule, $V_A = \frac{V_o R_1}{R_1 + R_2}$. By virtual short, $V_A = V_B$:

$$\frac{V_o R_1}{R_1 + R_2} = V_B \qquad \Rightarrow \qquad V_o = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{V_1 + V_2 + V_3}{3}\right)$$

The circuit performs addition of the three inputs and this sum is amplified by a factor $\left[(1 + R_2/R_1) \times \frac{1}{3}\right]$. If $R_2 = 2R_1$, then $V_o = V_1 + V_2 + V_3$ — a **non-inverting type adder**.

10. Adder / Subtractor



KCL at node A:

$$\frac{V_1 - V_A}{R} + \frac{V_2 - V_A}{R} + \frac{V_o - V_A}{R} = 0 \qquad \Rightarrow \qquad V_A = \frac{V_1 + V_2 + V_o}{3} \qquad (1)$$

KCL at node B:

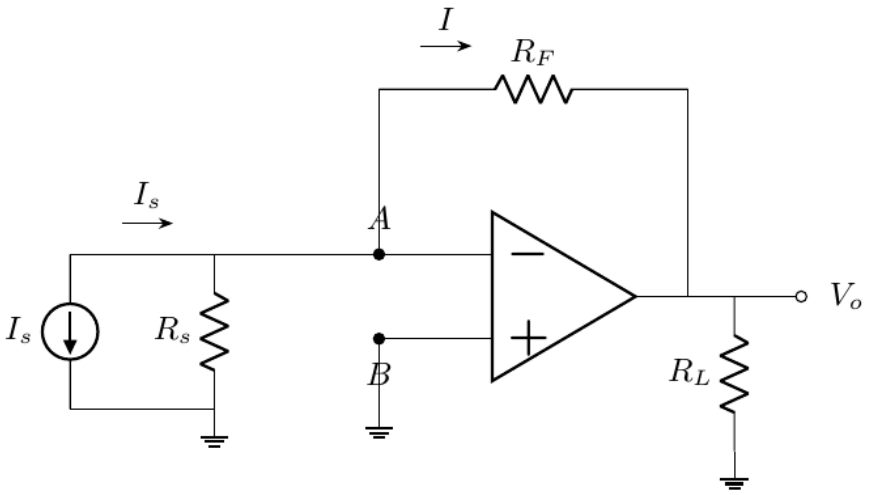
$$\frac{V_3 - V_B}{R} + \frac{V_4 - V_B}{R} + \frac{0 - V_B}{R} = 0 \qquad \Rightarrow \qquad V_B = \frac{V_3 + V_4}{3} \qquad (2)$$

By the virtual short concept, $V_A = V_B$:

$$\frac{V_1 + V_2 + V_o}{3} = \frac{V_3 + V_4}{3} \qquad \Rightarrow \qquad V_o = (V_3 + V_4) - (V_1 + V_2)$$

11. Current to Voltage Converter

An I-to-V converter is used to convert the output current of a transducer or sensor into a **proportionate voltage**. It is also called a **transresistance amplifier**. Example: the photocurrent of a **photodiode** can be converted into a proportionate voltage by using an I-to-V converter.



By virtual ground, $V_A = V_B = 0$.

The current through R_s will be **zero**, because both terminals of R_s are at zero potential.

KCL at node A:

$$I_s + I = 0 \quad \Rightarrow \quad I_s = -I = -\frac{V_o - 0}{R_F}$$

$$V_o = -I_s R_F \quad \Rightarrow \quad V_o \propto I_s \qquad \frac{V_o}{I_s} = -R_F \quad \text{(transresistance gain)}$$

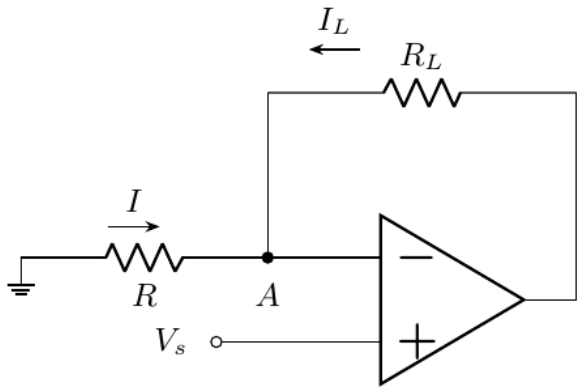
The output voltage V_o is **independent of the source and load impedance**.

12. Voltage to Current Converter

A V-to-I converter converts the output voltage of a transducer into a **proportionate current**. It is also called a **transconductance amplifier**. It is of two types: (i) **floating load** and (ii) **grounded load**.

(i) Floating load V-to-I converter

If no terminal of R_L is connected to ground, then it is called a **floating load**.



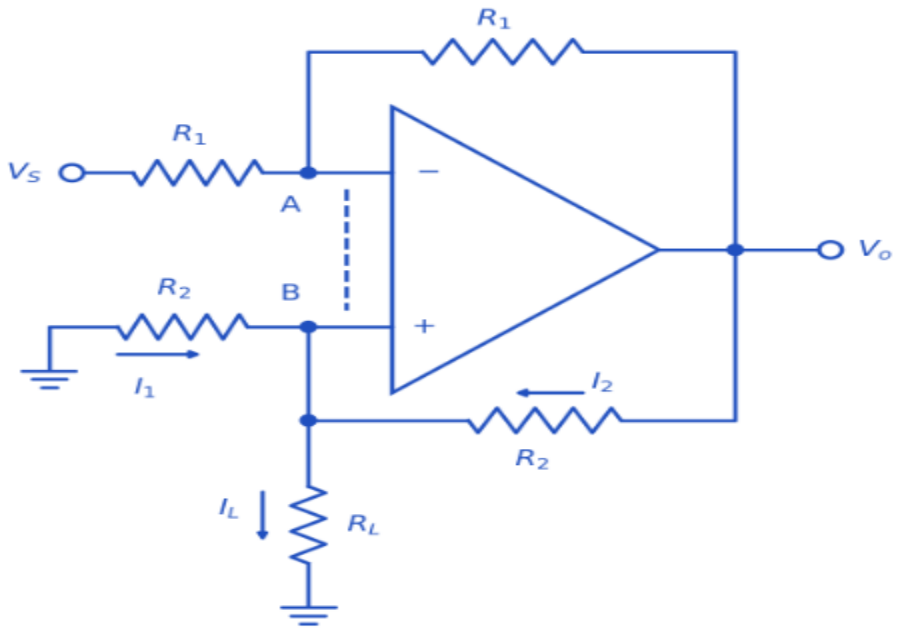
By the virtual short concept, $V_A = V_s$. KCL at node A:

$$I + I_L = 0 \quad \Rightarrow \quad \frac{0 - V_s}{R} + I_L = 0 \quad \Rightarrow \quad I_L = \frac{V_s}{R}$$

$$\frac{I_L}{V_s} = \frac{1}{R} \quad \text{(transconductance gain)}$$

The load current is **independent of source and load impedance**. If instead V_s is applied through R to the inverting terminal and the non-inverting terminal is grounded, then by virtual ground $V_A = 0$ and KCL gives $\frac{V_s}{R} + I_L = 0$, i.e. $I_L = \frac{-V_s}{R}$.

(ii) Grounded load V-to-I converter



Both **positive and negative feedback** are present. But due to the presence of negative feedback the op-amp remains in the linear region, and hence the virtual short concept can be used.

KCL at node B:

$$I_1 + I_2 = I_L \qquad \Rightarrow \qquad \frac{0 - V_B}{R_2} + \frac{V_o - V_B}{R_2} = I_L$$

$$V_o - 2V_B = I_L R_2 \qquad \Rightarrow \qquad V_B = \frac{V_o - I_L R_2}{2} \qquad (1)$$

By superposition at node A:

$$V_A = \frac{V_s R_1 + V_o R_1}{R_1 + R_1} = \frac{V_s + V_o}{2} \qquad (2)$$

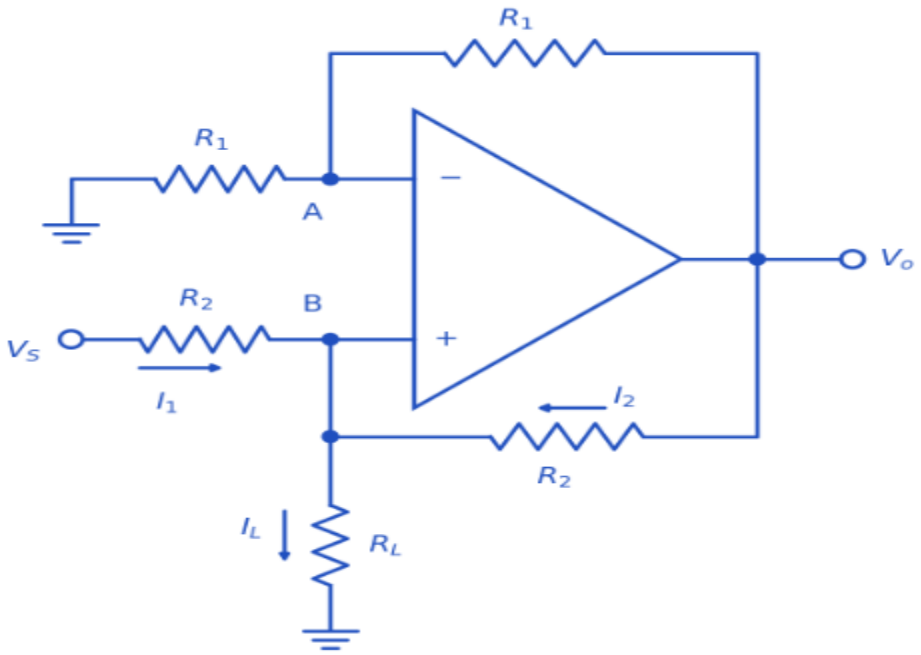
By the virtual short concept, $V_A = V_B$:

$$\frac{V_o - I_L R_2}{2} = \frac{V_s + V_o}{2} \qquad \Rightarrow \qquad I_L = \frac{-V_s}{R_2}$$

I_L does not depend on R_1 (nor on the source resistance R_s, if present), and also does not depend on R_L.

Worked problem — load current

Q. Find the load current in the circuit shown.



Solution.

$$V_A = \frac{V_o \times R_1}{2R_1} = \frac{V_o}{2} \qquad \text{and by virtual short} \qquad V_A = V_B = \frac{V_o}{2}$$

KCL at node B:

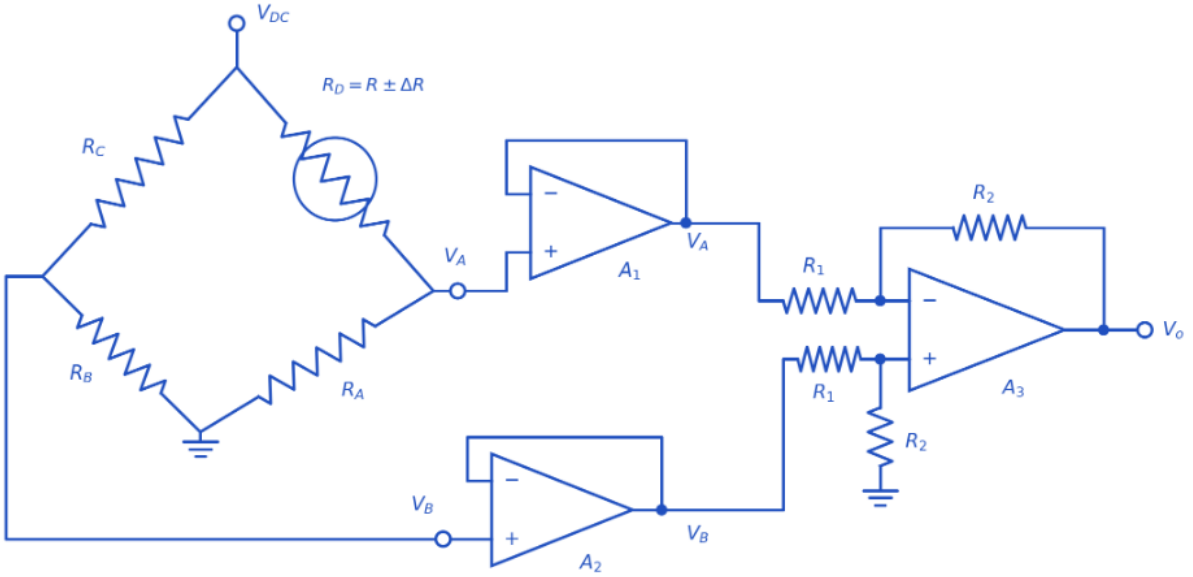
$$I_1 + I_2 = I_L \qquad \Rightarrow \qquad \frac{V_s - V_B}{R_2} + \frac{V_o - V_B}{R_2} = I_L$$

$$\frac{V_s}{R_2} + \frac{V_o}{R_2} - \frac{2V_B}{R_2} = I_L \qquad \Rightarrow \qquad I_L = \frac{V_s}{R_2}$$

The load current does not depend on the load resistance, and also does not depend on R_1 (or on the source resistance R_s, if present).

13. Instrumentation Amplifier

A circuit which converts a change in a **physical signal** into a proportionate **amplified voltage** is called an instrumentation amplifier. For example, a change in temperature can be converted into an e.m.f. or voltage. It consists of (i) a **transducer bridge**, (ii) **two voltage followers** and (iii) a **difference amplifier**.



- R_D is a **transducer** whose resistance can vary according to a change in the physical signal. For example, a thermistor or sensistor can be used in place of R_D to convert a change in temperature into a voltage.
- Op-amps A_1 and A_2 are **voltage followers**. They prevent current flowing out of the transducer bridge, i.e. they prevent **loading of the transducer bridge**. If the followers are not used, then R_1 and R_2 will draw current from the bridge, which disturbs the operation of the bridge.

- Op-amp A_3 is a **difference amplifier**: $V_o = \frac{R_2}{R_1}(V_B - V_A)$

Operation

When the transducer bridge is in the **balanced condition**, $V_A = V_B$, and the instrumentation amplifier generates **zero output**. If a change occurs in the physical signal, then R_D will change by $\pm\Delta R$; thus V_A changes. Hence V_A and V_B become unequal and a **non-zero output** is generated.

Calculation of V_o

Let $R_A = R_B = R_C = R$. Using voltage division:

$$V_B = \frac{V_{DC} \times R_B}{R_B + R_C} = \frac{V_{DC}}{2}$$

$$V_A = \frac{V_{DC} \times R_A}{R_A + R_D} = \frac{V_{DC} \times R}{2R \pm \Delta R}$$

Now,

$$V_o = \frac{R_2}{R_1}(V_B - V_A) = \frac{R_2}{R_1} \times V_{DC}\left[\frac{1}{2} - \frac{R}{2R \pm \Delta R}\right]$$

$$= \frac{V_{DC}R_2}{R_1}\left[\frac{2R \pm \Delta R - 2R}{2(2R \pm \Delta R)}\right]$$

$$= \frac{V_{DC}R_2}{R_1}\left[\frac{\pm\Delta R}{4R \pm 2\Delta R}\right]$$

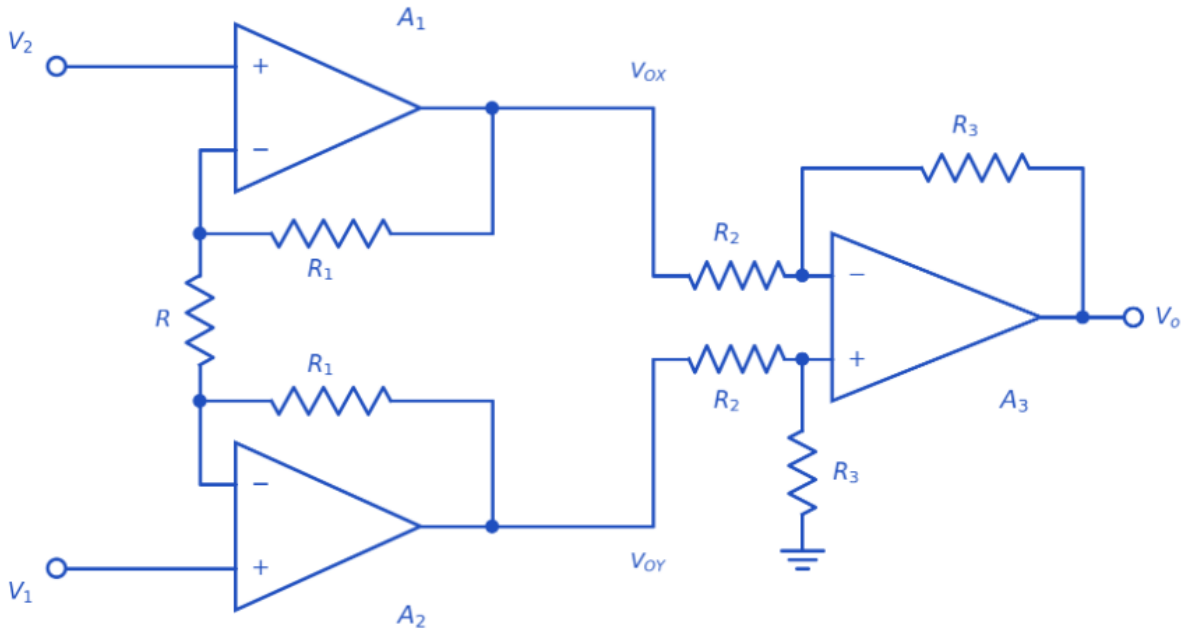
But $2\Delta R \lll 4R$, so:

$$V_o \cong \frac{V_{DC}R_2}{R_1}\left(\frac{\pm\Delta R}{4R}\right) \qquad \Rightarrow \qquad V_o \propto \Delta R$$

ΔR is proportional to the change in the physical signal.

14. Three Op-Amp Instrumentation Amplifier — Worked Problem

Q. For the op-amp circuit shown, find the output voltage.



Solution. V_o can be calculated by using the **superposition principle**.

Case 1 — V_1 applied, $V_2 = 0$, $V_o = V_{o1}$

A_2 becomes a non-inverting amplifier and A_1 becomes an inverting amplifier:

$$V_{oy} = \left(1 + \frac{R_1}{R}\right) V_1 \qquad V_{ox} = \left(\frac{-R_1}{R}\right) V_1$$

Op-amp A_3 is a difference amplifier:

$$V_{o1} = \frac{R_3}{R_2}(V_{oy} - V_{ox})$$

$$V_{o1} = \frac{R_3}{R_2}\left\{\left(1 + \frac{R_1}{R}\right) V_1 + \frac{R_1}{R} V_1\right\}$$

$$V_{o1} = \left[\frac{R_3}{R_2}\left(1 + \frac{2R_1}{R}\right)\right] V_1 \qquad (1)$$

Case 2 — V_2 applied, $V_1 = 0$, $V_o = V_{o2}$

A_1 becomes a non-inverting amplifier and A_2 becomes an inverting amplifier:

$$V_{ox} = \left(1 + \frac{R_1}{R}\right) V_2 \qquad V_{oy} = \frac{-R_1}{R} V_2$$

$$V_{o2} = \frac{R_3}{R_2}\{V_{oy} - V_{ox}\}$$

$$V_{o2} = \frac{-R_3}{R_2}\left\{\left(1 + \frac{2R_1}{R}\right)V_2\right\} \qquad (2)$$

Net output

If both V_1 and V_2 are applied, the net output is $V_o = V_{o1} + V_{o2}$:

$$V_o = \frac{R_3}{R_2}\left(1 + \frac{2R_1}{R}\right)(V_1 - V_2)$$

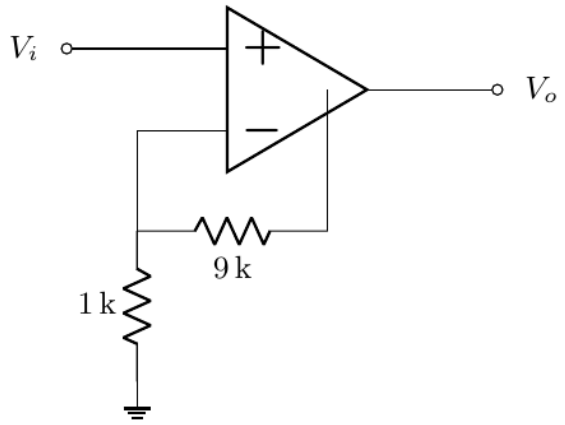
$$\text{Gain} = \frac{V_o}{V_1 - V_2} = \frac{R_3}{R_2}\left(1 + \frac{2R_1}{R}\right)$$

The above circuit is also used as an instrumentation amplifier if the voltages V_A and V_B obtained from the transducer bridge are connected as V_1 and V_2. It provides **better gain** than the previous instrumentation amplifier, because op-amps A_1 and A_2 also contribute a gain of $(1 + 2R_1/R)$.

15. Further Worked Problems

Q4 — 3 dB bandwidth

Q. In the circuit shown, the op-amp has a gain $\frac{10^4}{1+10^{-3}s}$. Calculate the 3 dB bandwidth in rad/sec for the amplifier shown.



Solution. Gain of the op-amp:

$$A_V = \frac{10^4}{1 + 10^{-3}s} = \frac{10^4}{1 + s/1000} = \frac{10^4}{1 + j\omega/1000} = \frac{A_{OL}}{1 + j\omega/\omega_{OL}}$$

$$A_{OL} = 10^4 \qquad \omega_{OL} = 1000 \text{ rad/sec} \qquad \text{UGB} = A_{OL} \times \omega_{OL} = 10^7 \text{ rad/sec}$$

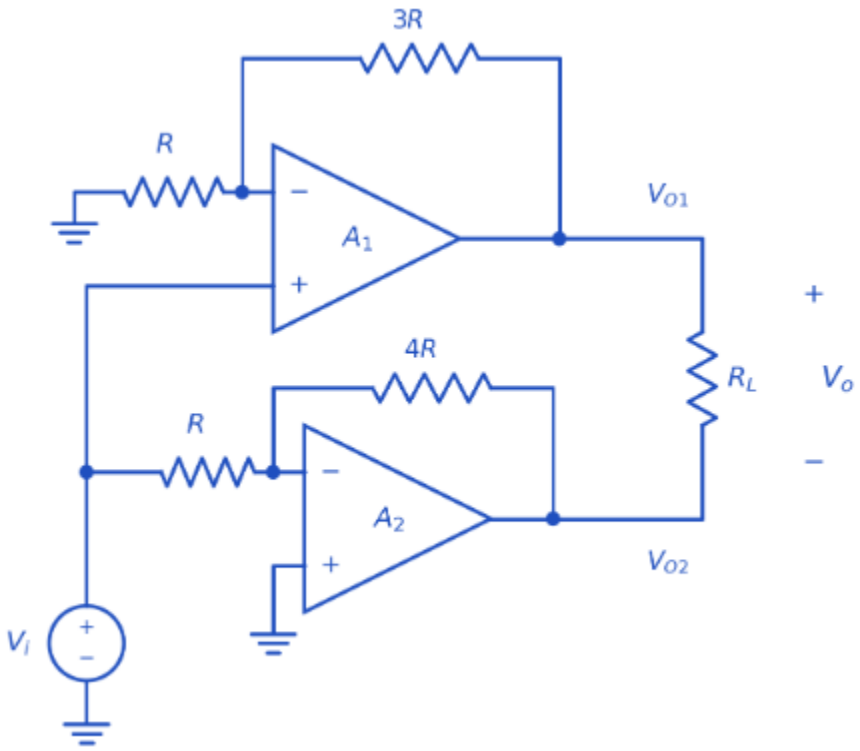
For the non-inverting amplifier:

$$A_{CL} = 1 + \frac{R_F}{R} = 1 + \frac{9}{1} = 10$$

$$(BW)_f = \frac{\text{UGB}}{A_{CL}} = \frac{10^7}{10} = 10^6 \text{ rad/sec}$$

Q5 — voltage gain of a two op-amp circuit

Q. Calculate the voltage gain V_o/V_i in the op-amp circuit shown.



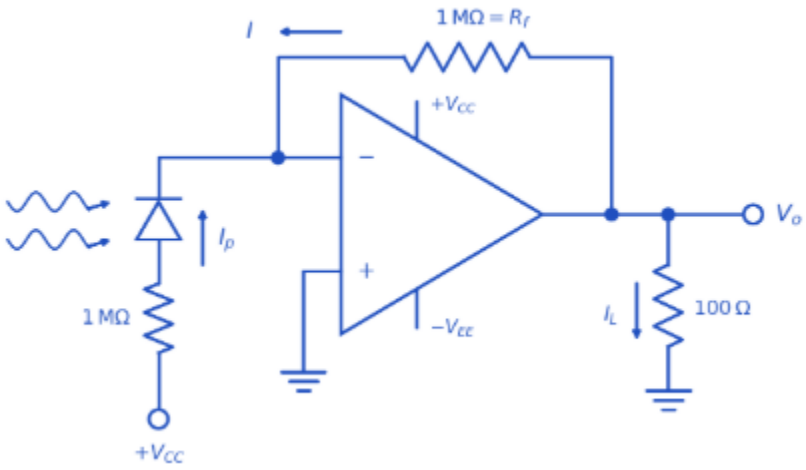
Solution. A_1 is a non-inverting amplifier and A_2 is an inverting amplifier:

$$V_{o1} = \left(1 + \frac{3R}{R}\right) V_i = 4V_i \qquad V_{o2} = \frac{-4R}{R} V_i = -4V_i$$

$$V_o = V_{o1} - V_{o2} = 4V_i - (-4V_i) = 8V_i \quad \Rightarrow \quad \frac{V_o}{V_i} = 8$$

Q6 — photodiode I-to-V converter

Q. In the circuit shown, the op-amp is ideal and the photodiode has a responsivity of 0.8 A/W. If 10 μW of optical power is incident on the photodiode, calculate the load current.



Solution.

$$\text{Responsivity} = \frac{\text{photocurrent}}{\text{optical power}} = \frac{I_p}{P_{opt}} \qquad \Rightarrow \qquad I_p = 0.8 \times 10 \times 10^{-6} = 8\ \mu\text{A}$$

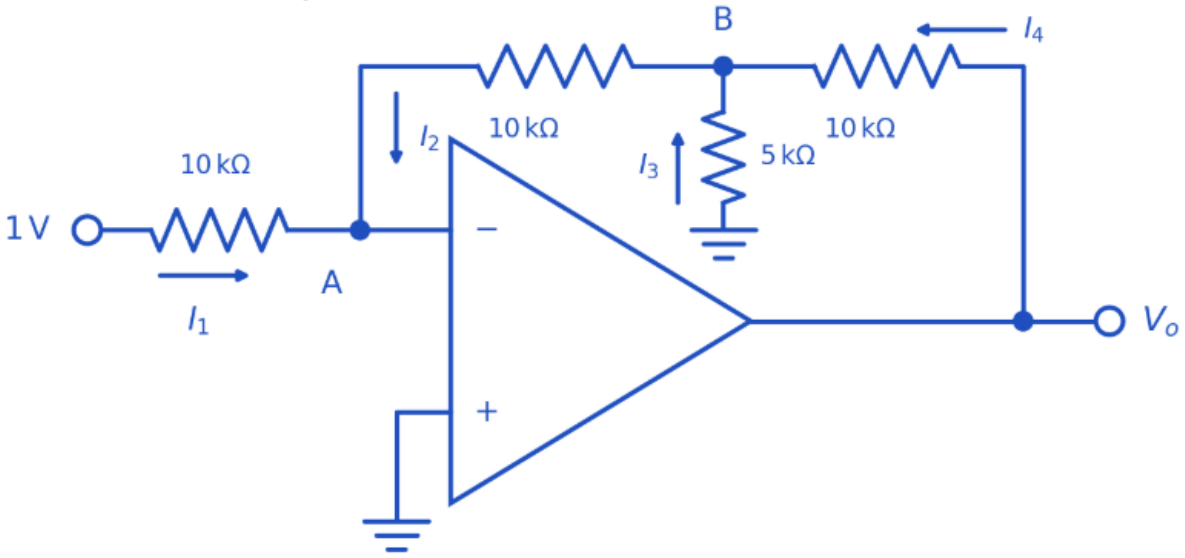
KCL at node A:

$$I_p + I = 0 \qquad \Rightarrow \qquad I_p + \frac{V_o - 0}{R_F} = 0 \qquad \Rightarrow \qquad V_o = -I_p R_F = -8\ \text{V}$$

$$I_L = \frac{V_o}{100} = \frac{-8}{100} = -80\ \text{mA}$$

Q7 — T-network feedback

Q. Calculate the output voltage in the op-amp circuit shown.



Solution. $V_A = 0$ (virtual ground).

KCL at node A:

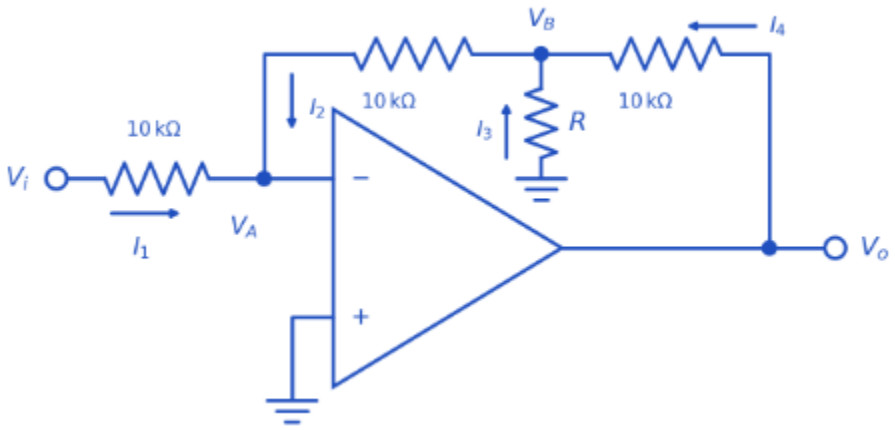
$$I_1 + I_2 = 0 \quad \Rightarrow \quad \frac{1}{10} + \frac{V_B - 0}{10} = 0 \quad \Rightarrow \quad V_B = -1\ \text{V}$$

KCL at node B:

$$I_3 + I_4 = I_2 \quad \Rightarrow \quad \frac{0 - V_B}{5} + \frac{V_o - V_B}{10} = \frac{V_B - 0}{10}$$

$$\frac{-2V_B}{10} + \frac{V_o}{10} - \frac{V_B}{10} = \frac{V_B}{10} \quad \Rightarrow \quad V_o = 4V_B = 4 \times (-1) = -4\ \text{V}$$

Q. 8 For the circuit shown, calculate the resistance R such that the gain V_o/V_i is -10.



Solution. By the virtual ground concept, $V_A = 0$.

KCL at node A:

$$I_1 + I_2 = 0 \quad \Rightarrow \quad \frac{V_i - 0}{10} + \frac{V_B - 0}{10} = 0 \quad \Rightarrow \quad V_i = -V_B$$

KCL at node B:

$$I_3 + I_4 = I_2 \quad \Rightarrow \quad \frac{0 - V_B}{R} + \frac{V_o - V_B}{10} = \frac{V_B - 0}{10}$$

Substituting $V_B = -V_i$:

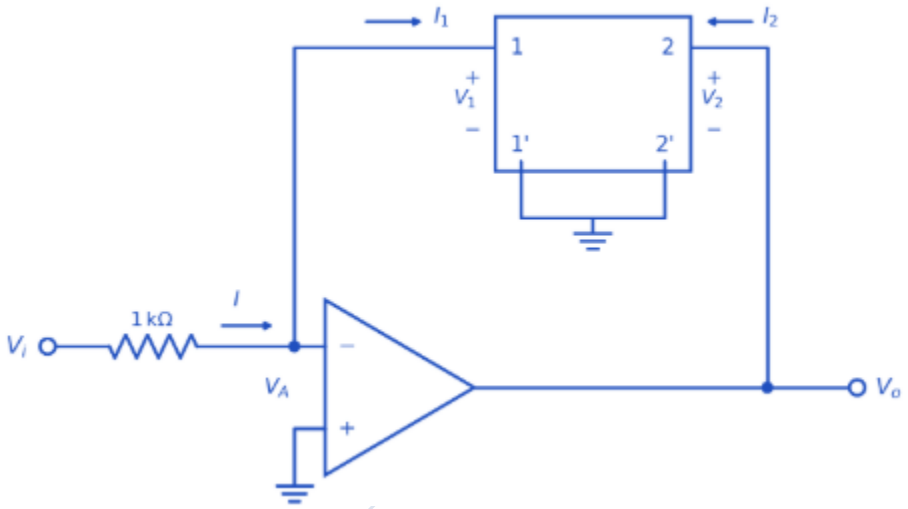
$$\frac{V_i}{R} + \frac{V_o + V_i}{10} = \frac{-V_i}{10} \quad \Rightarrow \quad V_i\left[\frac{1}{R} + \frac{2}{10}\right] = \frac{-V_o}{10}$$

$$\frac{V_o}{V_i} = -10\left[\frac{1}{R} + \frac{1}{5}\right] = -10 \qquad \Rightarrow \qquad \frac{1}{R} = 1 - \frac{1}{5} = \frac{4}{5}$$

$$R = \frac{5}{4} = 1.25\ \text{k}\Omega$$

Q9 — feedback network specified by z-parameters

Q. Calculate the voltage gain in the op-amp circuit shown, if the z-parameters for the feedback network are $Z_{11} = Z_{22} = 11$ k and $Z_{12} = Z_{21} = 1$ k.



Solution. The z-parameter equations are:

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \;\Rightarrow\; V_1 = 11I_1 + I_2 \qquad (1)$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \;\Rightarrow\; V_2 = I_1 + 11I_2 \qquad (2)$$

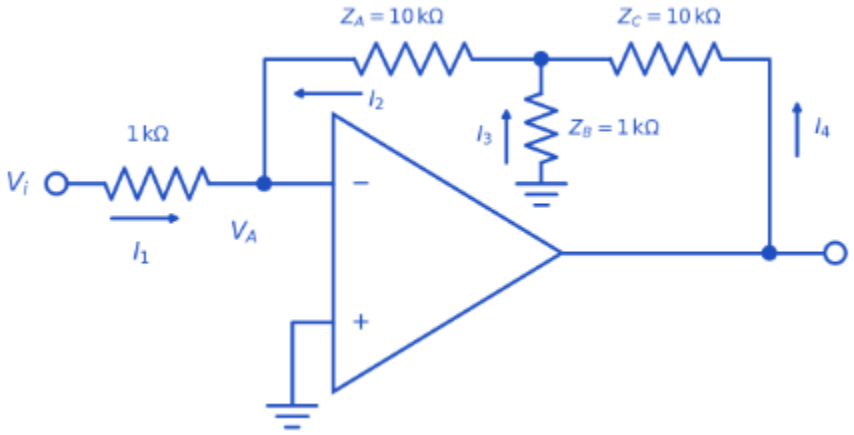
From the circuit, $V_1 = V_A = 0$ and $V_2 = V_o$. KCL at node A: $I = I_1$, and $\frac{V_i - 0}{1} = I_1$, so $I_1 = V_i$.

From equation (1): $0 = 11I_1 + I_2 \Rightarrow I_2 = -11V_i$. From equation (2):

$$V_o = V_i + 11 \times (-11V_i) = V_i - 121V_i \qquad \Rightarrow \qquad \frac{V_o}{V_i} = -120$$

Method 2 — T-equivalent of the feedback network

From the given z-parameters the network is **reciprocal and symmetrical**, so it can be redrawn as a T-network:



$$Z_a = Z_{11} - Z_{12} = 10\,\text{k} \qquad Z_b = Z_{12} = 1\,\text{k} \qquad Z_c = Z_{22} - Z_{12} = 10\,\text{k}$$

KCL at node A:

$$I_1 + I_2 = 0 \quad \Rightarrow \quad \frac{V_i}{1} + \frac{V_B}{10} = 0 \quad \Rightarrow \quad V_B = -10V_i$$

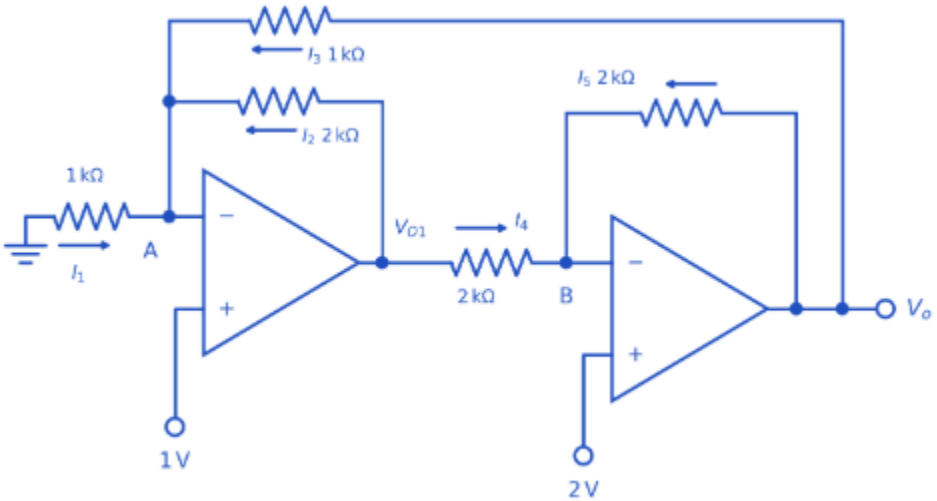
KCL at node B:

$$\frac{0 - V_B}{1} + \frac{V_o - V_B}{10} = \frac{V_B}{10} \quad \Rightarrow \quad V_o = 12V_B = -120V_i$$

$$\frac{V_o}{V_i} = -120$$

Q10 — two-stage circuit

Q. Find V_o for the circuit shown, given inputs of 1 V and 2 V.



Solution. $V_A = 1$ V and $V_B = 2$ V (virtual short at each stage).

KCL at node A:

$$I_1 + I_2 + I_3 = 0 \qquad \Rightarrow \qquad \frac{0-1}{2} + \frac{V_{o1}-1}{2} + \frac{2-1}{1} = 0$$

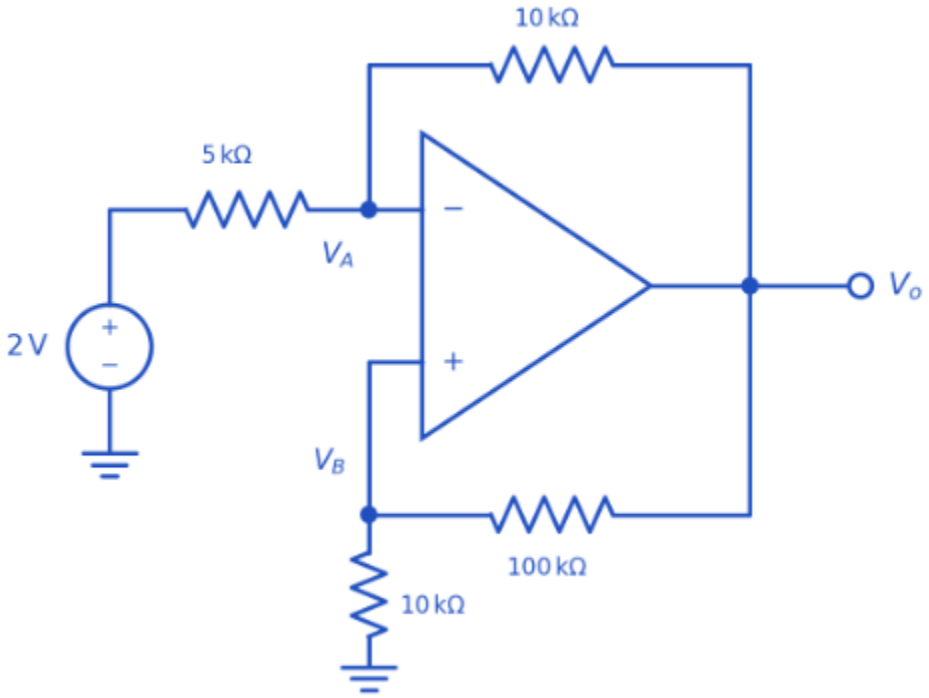
$$-1 + V_{o1} - 1 + 2 = 0 \qquad \Rightarrow \qquad V_{o1} = 0$$

KCL at node B:

$$I_4 + I_5 = I_3 \qquad \Rightarrow \qquad \frac{0-2}{2} + \frac{V_o - 2}{2} = \frac{2-1}{1} \qquad \Rightarrow \qquad V_o = 6 \text{ volts}$$

Q11 — find the output voltage

Q. Find the output voltage V_o for the circuit shown.



Solution. By superposition at node A, and voltage division at node B:

$$V_A = \frac{2 \times 10 + V_o \times 5}{15} \qquad V_B = \frac{V_o \times 10}{10 + 100}$$

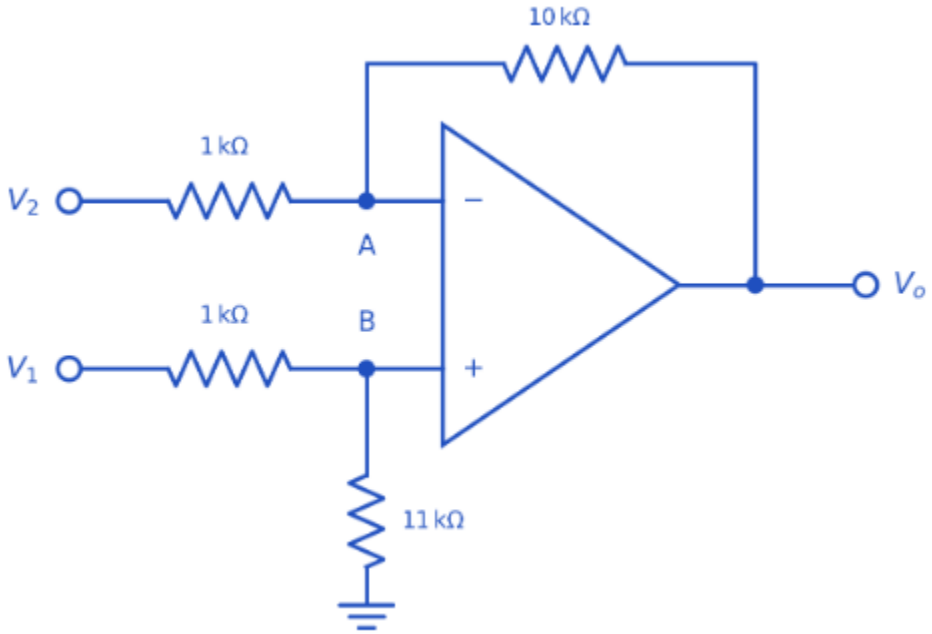
By virtual short, $V_A = V_B$:

$$\frac{20 + 5V_o}{15} = \frac{10V_o}{110} \qquad \Rightarrow \qquad \frac{4 + V_o}{3} = \frac{V_o}{11} \qquad \Rightarrow \qquad 44 + 11V_o = 3V_o$$

$$V_o = \frac{-44}{8} = -5.5 \text{ V}$$

Q12 — CMRR of a non-ideal difference amplifier

Q. For the op-amp circuit shown, calculate the CMRR.



Solution. The given circuit is a difference amplifier in which a **mismatch is present in the value of** R_2.

EQUATION (1)

1 Describe the two inputs in a new way

Difference-mode: $V_d = V_1 - V_2$ Common-mode (average): $V_{CM} = \frac{V_1 + V_2}{2}$

Going back: $V_1 = V_{CM} + \frac{V_d}{2}$ $V_2 = V_{CM} - \frac{V_d}{2}$

Example: $V_1 = 5.01\,\text{V}$, $V_2 = 4.99\,\text{V} \Rightarrow V_d = 0.02\,\text{V}$, $V_{CM} = 5\,\text{V}$
"Both inputs sit at 5 V (common), plus a tiny ±0.01 V difference" ($5 + 0.01 = 5.01$, $5 - 0.01 = 4.99$)

2 Linear amplifier → superposition → one gain for each part

$$V_o = A_{DM} V_d + A_{CM} V_{CM} \quad \cdots(1)$$

A_{DM} = gain for the WANTED signal (difference) A_{CM} = gain for the UNWANTED signal (common level)

Ideal amplifier: $A_{CM} = 0$ (ignores the 5 V, amplifies only 0.02 V). Real amplifier: small $A_{CM} \neq 0$

EQUATION (2)

1 Linear circuit → output is a weighted sum of inputs

$$V_o = A_1 V_1 + A_2 V_2$$

A_1 = gain seen by V_1 (set $V_2 = 0$: $A_1 = V_o/V_1$) $\qquad$ A_2 = gain seen by V_2 (set $V_1 = 0$: $A_2 = V_o/V_2$)

2 Substitute V_1 and V_2 in terms of V_d and V_{CM}

$$V_o = A_1\left(V_{CM} + \frac{V_d}{2}\right) + A_2\left(V_{CM} - \frac{V_d}{2}\right)$$

3 Expand the brackets

$$V_o = A_1 V_{CM} + \frac{A_1}{2} V_d + A_2 V_{CM} - \frac{A_2}{2} V_d$$

4 Group V_d terms and V_{CM} terms

$$V_o = \left(\frac{A_1 - A_2}{2}\right) V_d + (A_1 + A_2)\, V_{CM} \qquad \cdots(2)$$

COMPARE (1) and (2) — same output, so coefficients must match

$$A_{DM} = \frac{A_1 - A_2}{2} \qquad A_{CM} = A_1 + A_2$$

$A_{CM} = A_1 + A_2$ measures IMBALANCE (A_2 is negative since V_2 goes to the − input).

$A_{DM} = \frac{A_1 - A_2}{2}$ = average strength with which the inputs push the output in opposite directions.

Ideal ($R_2 = 10\,\text{k}\Omega$)

$A_1 = 10, \quad A_2 = -10$

$A_{DM} = \frac{10 + 10}{2} = 10$

$A_{CM} = 10 - 10 = 0 \Rightarrow \text{CMRR} = \infty$

Mismatch ($R_2 = 11\,\text{k}\Omega$)

$A_1 = \frac{121}{12} \approx 10.083, \quad A_2 = -10$

$A_{DM} = \frac{241}{24} \approx 10.04$

$A_{CM} = \frac{1}{12} \approx 0.083 \Rightarrow \text{CMRR} = 120.5$

By virtual short, $V_A = V_B$:

$$\frac{V_o \times 1 + 10V_2}{11} = \frac{11 \times V_1}{12} \qquad \Rightarrow \qquad V_o = \frac{121V_1}{12} - 10V_2$$

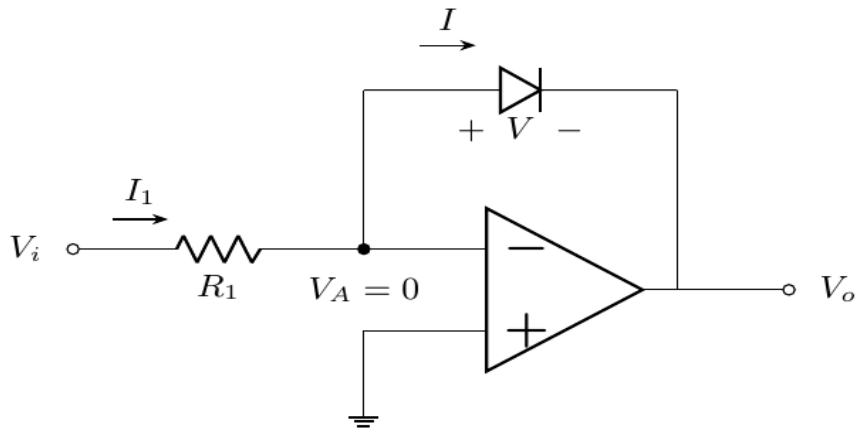
$$A_1 = \frac{121}{12} \qquad A_2 = -10$$

$$A_{DM} = \frac{\frac{121}{12} + 10}{2} = \frac{241}{24} \qquad A_{CM} = \frac{121}{12} - 10 = \frac{1}{12}$$

$$CMRR = \frac{A_{DM}}{A_{CM}} = \frac{241}{24} \times 12 = 120.5$$

2. Log Amplifier

A log amplifier provides an output voltage which is proportional to the **logarithm** of the input voltage.



Since $V_A = 0$, the voltage across the diode is $V = V_A - V_o = -V_o$.

KCL at node A: $I_1 = I$

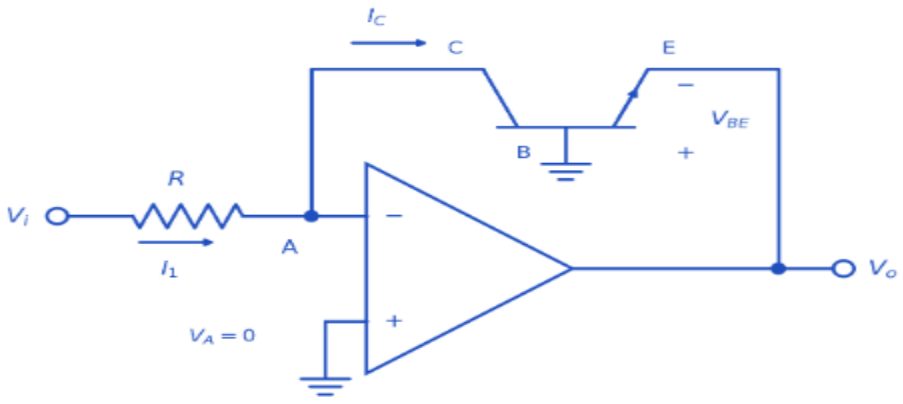
$$\frac{V_i - 0}{R} = I_0 e^{V/\eta V_T} \qquad \Rightarrow \qquad \frac{V_i}{I_0 R} = e^{V/\eta V_T} = e^{-V_o/\eta V_T}$$

Taking the logarithm on both sides:

$$\ln\left(\frac{V_i}{I_0 R}\right) = \frac{-V_o}{\eta V_T} \qquad \Rightarrow \qquad V_o = -\eta V_T \ln\left(\frac{V_i}{I_0 R}\right)$$

$$V_o \propto \ln(K V_i)$$

Log amplifier using a BJT



A **BJT** can also be used in place of the diode. Both the collector and base are at 0 V, hence collector and base are **virtually shorted**, so the BJT acts as a **diode**.

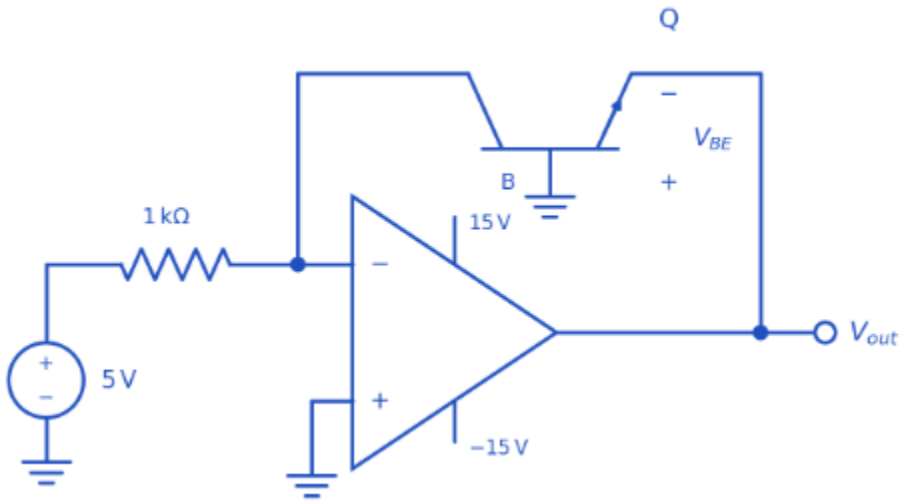
$$V_{BE} = V_B - V_E = 0 - V_o \qquad \Rightarrow \qquad V_{BE} = -V_o$$

KCL at node A**:** $I_1 = I_c$

$$\frac{V_i - 0}{R} = I_S e^{V_{BE}/V_T} \qquad \Rightarrow \qquad \frac{V_i}{I_S R} = e^{-V_o/V_T}$$

$$V_o = -V_T \ln\left(\frac{V_i}{I_S R}\right) \qquad \Rightarrow \qquad V_o \propto \ln(KV_i)$$

Q1 — output voltage with a silicon transistor



Q. In the circuit shown, what is the output voltage V_{out} if a silicon transistor Q and an ideal op-amp are used? (a) -15 V (b) -0.7 V (c) 0.7 V (d) $+15$ V

Solution. Since the saturation current is not given, the actual value of V_o cannot be calculated. So $V_o = -V_{BE} = -0.7$ V.

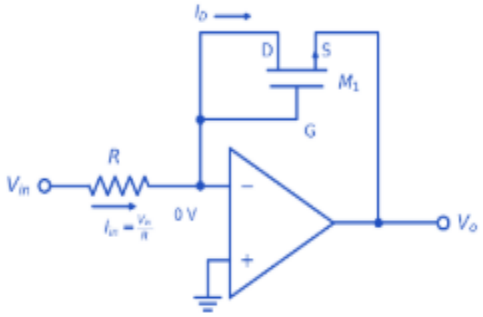
Q2 — repeat with $I_S = 5 \times 10^{-14}$ A

$$V_o = -V_T \ln\left(\frac{V_i}{I_S R}\right) = -26\ln\left(\frac{5}{5 \times 10^{-14} \times 1 \times 10^{3}}\right) \text{ mV} = -26\ln(10^{11}) \text{ mV}$$

$$V_o = -0.658 \text{ volts}$$

Op-Amp with MOSFET in Feedback: Square-Root or Log?

Diode-connected NMOS in the feedback path



Common to both cases:

Virtual ground ⇒ $I_D = I_{in} = \frac{V_{in}}{R}$

Gate = Drain = 0 V, Source = V_o

⇒ $V_{GS} = V_{DS} = -V_o$

Diode-connected ⇒ always in saturation ($V_{DS} = V_{GS} > V_{GS} - V_{th}$)

Result depends on the MOSFET region (i.e. on how much current I_D flows):

CASE 1: Strong inversion (default)

normal current levels — textbook assumption

Square law:

$$\frac{V_{in}}{R} = \frac{k}{2}(V_{GS} - V_{th})^2$$

$$V_{GS} = -V_o \Rightarrow -V_o - V_{th} = \sqrt{\frac{2V_{in}}{kR}}$$

$$V_o = -\left(V_{th} + \sqrt{\frac{2V_{in}}{kR}}\right)$$

SQUARE-ROOT AMPLIFIER

$(|V_o| - V_{th}) \propto \sqrt{V_{in}}$

(note: $V_o \propto \sqrt{V_{in}}$ only if V_{in} is ignored)

CASE 2: Weak inversion (subthreshold)

very small current (pA – sub-μA), large R

Exponential law:

$$\frac{V_{in}}{R} = I_0 e^{V_{GS}/(nV_T)}$$

$$V_{GS} = -V_o \Rightarrow -\frac{V_o}{nV_T} = \ln\frac{V_{in}}{I_0 R}$$

$$V_o = -nV_T \ln\left(\frac{V_{in}}{I_0 R}\right)$$

LOG AMPLIFIER

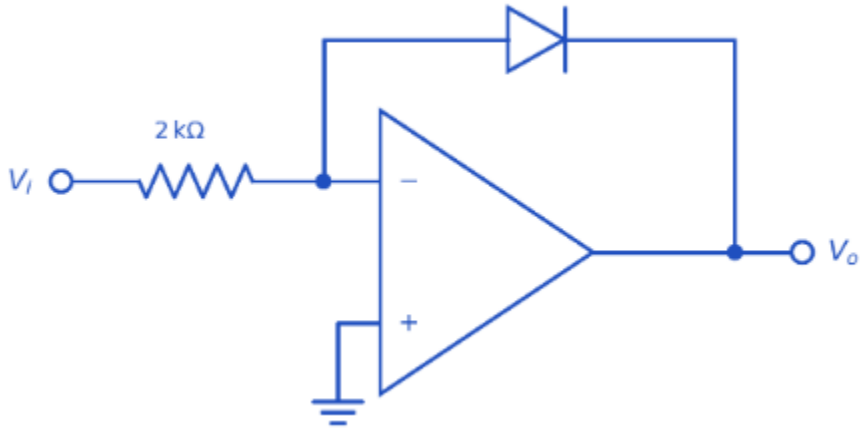
$V_o \propto \ln V_{in}$ (≈ 70–90 mV/decade)

special design case

If a **MOSFET** (Strong inversion) is used in the feedback path, then the circuit will act as a **square root amplifier**, i.e. $V_o \propto \sqrt{V_i}$.

Q3 — relationship between two outputs

Q. In the op-amp circuit shown, assume the diode current follows $I = I_S \exp(V/V_T)$. For $V_i = 2$ V, $V_o = V_{o1}$, and for $V_i = 4$ V, $V_o = V_{o2}$. The relationship between V_{o1} and V_{o2} is: (a) $V_{o2} = \sqrt{2}\,V_{o1}$ (b) $V_{o2} = e^2 V_{o1}$ (c) $V_{o2} = V_{o1}\ln 2$ (d) $V_{o1} - V_{o2} = V_T \ln 2$



Solution.

$$V_o = -V_T \ln\left(\frac{V_i}{I_S R}\right) = -V_T \ln(V_i) + V_T \ln(I_S R)$$

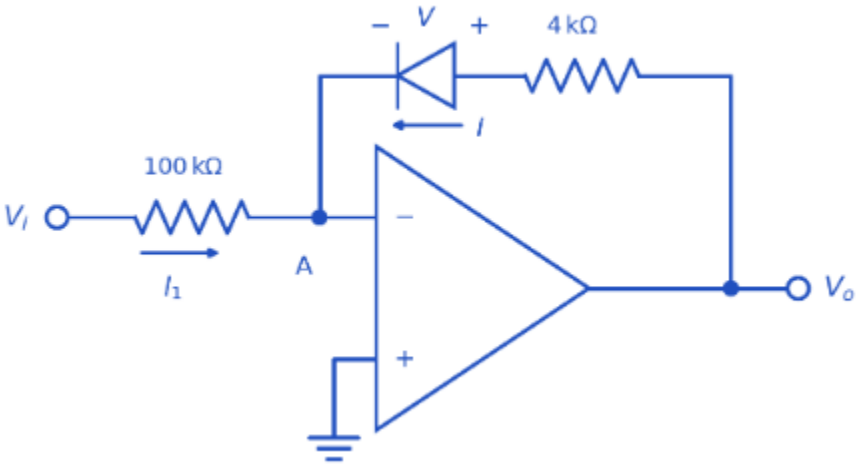
$$V_{o1} = -V_T \ln(2) + V_T \ln(I_S R)$$

$$V_{o2} = -V_T \ln(4) + V_T \ln(I_S R)$$

$$V_{o1} - V_{o2} = -V_T \ln(2) + V_T \ln(4) = V_T \ln\left(\frac{4}{2}\right) = V_T \ln(2) \qquad \text{— option (d)}$$

Q4 — diode with the $\left(e^{V/V_T} - 1\right)$ law

Q. In the circuit shown, the op-amp is ideal and the diode current follows $I = I_0\left(e^{V/V_T} - 1\right)$, where $I_0 = 1\ \mu\text{A}$ and $V_T = 25$ mV. Calculate the output voltage if $V_i = -1$ V.



Solution. $V_A = 0$.

KCL at node A:

$$I + I_1 = 0 \qquad \Rightarrow \qquad I = \frac{-V_i}{100\,\text{k}} = \frac{1}{100\,\text{k}} = 10\ \mu\text{A}$$

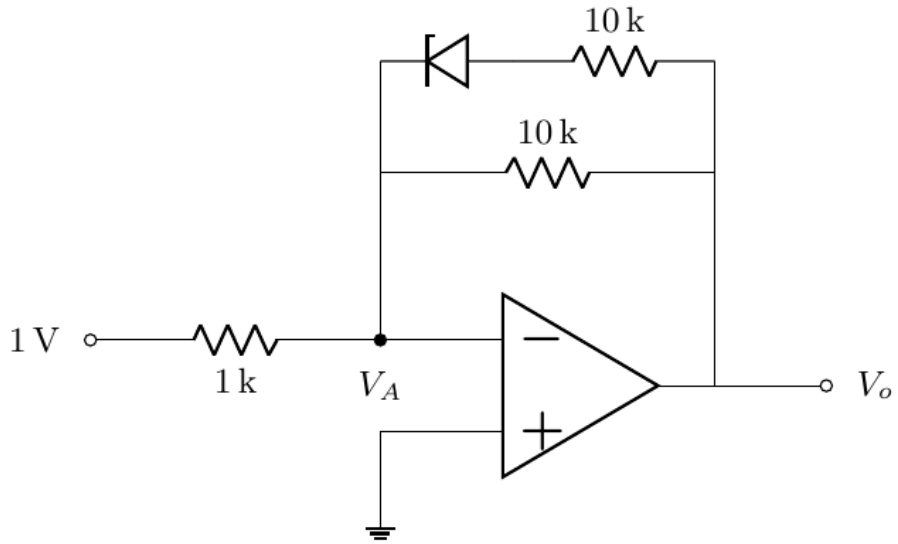
Also, from $I = I_0\left(e^{V/V_T} - 1\right)$:

$$V = V_T \ln\left(1 + \frac{I}{I_0}\right) = 0.025 \ln\left(1 + \frac{10}{1}\right) = 0.06 \text{ volts}$$

$$V_o - 0 = 4000\, I + V = 4000 \times 10 \times 10^{-6} + 0.06 = 0.04 + 0.06 = 0.1 \text{ V} = 100 \text{ mV}$$

Q5 — circuit with a zener diode

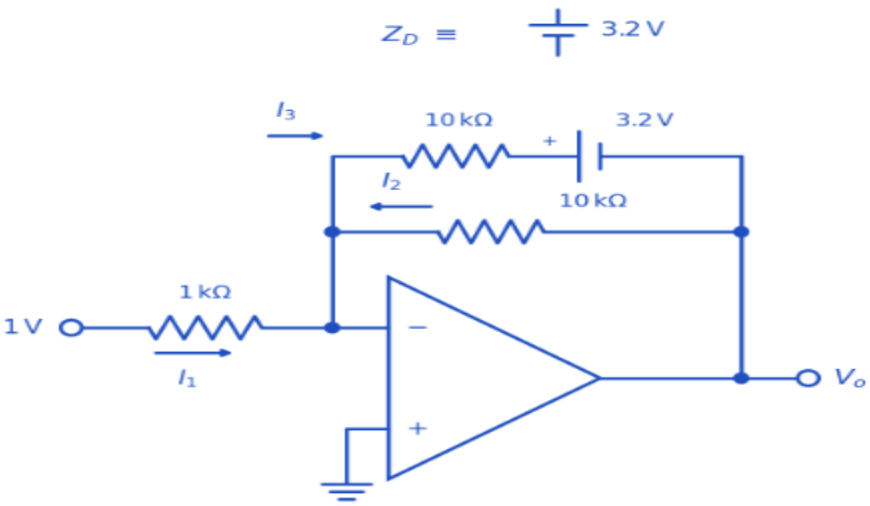
Q. In the circuit shown, the op-amp is ideal and the zener diode has a breakdown voltage of 3.2 V. Calculate the output voltage.



Solution. Let the zener diode be **off**; then the circuit is an inverting amplifier:

$$V_o = \frac{-R_F}{R} \times V_i = \frac{-10}{1} \times 1 = -10 \text{ volts}$$

Hence the zener diode is in **reverse bias** and also in the **breakdown region**, so it can be replaced with a 3.2 V source.



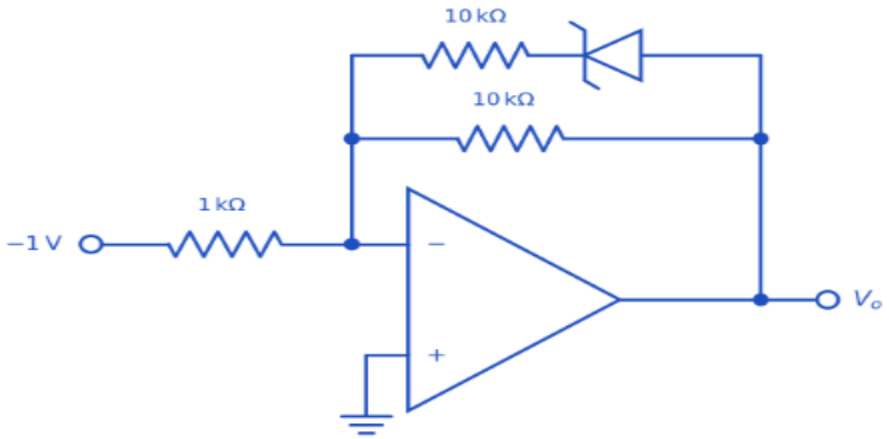
KCL at node A:

$$I_1 + I_2 = I_3 \quad \Rightarrow \quad \frac{1-0}{1} + \frac{V_o - 0}{10} = \frac{0 - 3.2 - V_o}{10}$$

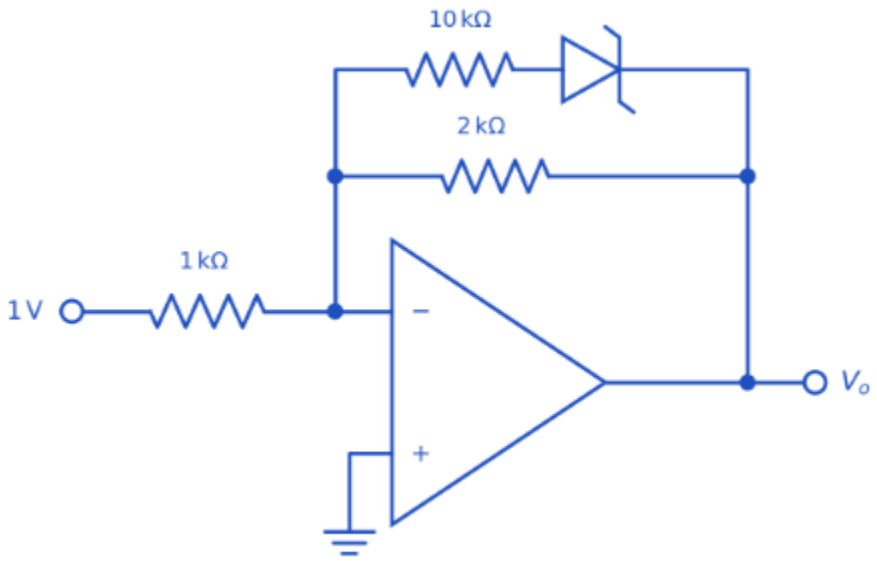
$$10 + V_o = -3.2 - V_o \quad \Rightarrow \quad 2V_o = -13.2 \quad \Rightarrow \quad V_o = -6.6 \text{ volts}$$

Two related cases.

- With $V_i = -1$ V, $R = 1$ k and a 10 k feedback shunted by the zener: let the zener be off, then $V_o = \frac{-10}{1} \times (-1) = 10$ V. Hence the zener is in **forward bias** and can be replaced with a short circuit, so both 10 k resistors come in parallel (5 k): $V_o = \frac{-5}{1} \times (-1) = 5$ V.

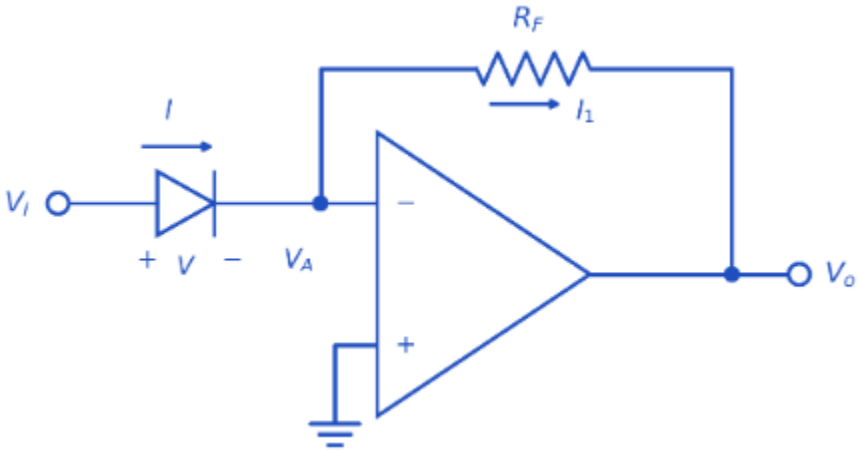


- With $V_i = 1$ V, $R = 1$ k and a 2 k feedback: let the zener be off, then $V_o = -2$ V. Hence the zener is reverse biased but **not in breakdown**, and acts as an open circuit: $V_o = -2$ V.



3. Antilog Amplifier (Exponential Amplifier)

An antilog amplifier provides an output voltage which is proportional to the **antilogarithm** of the input voltage.



$V_A = 0$, so $V = V_i - V_A = V_i$.

KCL at node A:

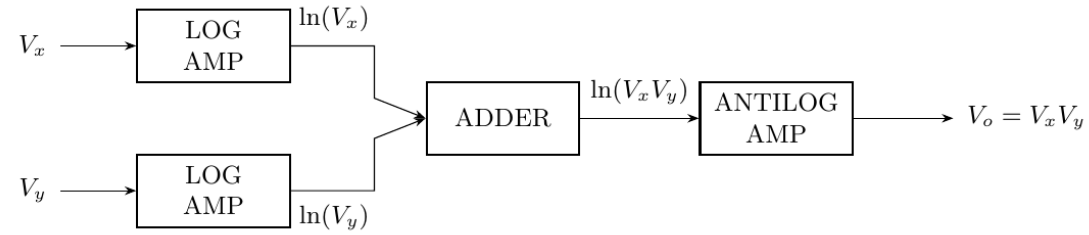
$$I + I_1 = 0 \qquad \Rightarrow \qquad I_0 e^{V_i/\eta V_T} = -\frac{V_o - 0}{R_F}$$

$$V_o = -R_F I_0 \, e^{V_i/\eta V_T} \qquad \Rightarrow \qquad V_o \propto e^{KV_i}$$

(e^x is the antilog of x.)

4. Multiplier

A circuit which can multiply two analog signals.





Symbol of Multiplier

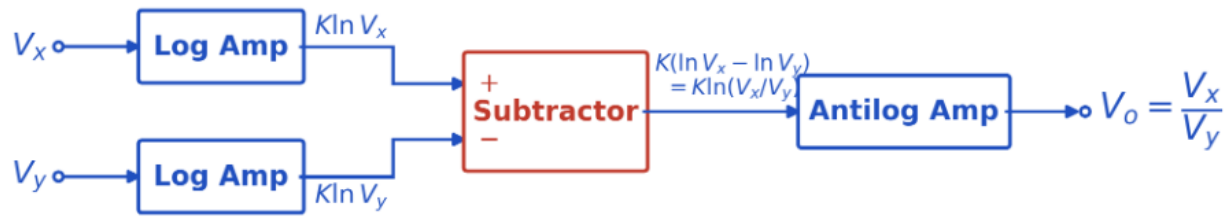
Why divide by V_{ref}?

1. $V_x \cdot V_y$ gives V^2, so dividing by V_{ref} brings the output back to volts and acts as the scale factor $K = \frac{1}{V_{ref}}$.
2. It keeps the output within the supply range: with $V_{ref} = 10$ V, full-scale inputs give $\frac{10 \times 10}{10} = 10$ V instead of 100.

If the adder is replaced with a **subtractor**, then the circuit becomes a **divider**.

Divider using Log / Antilog Amplifiers

If the adder is replaced with a subtractor, the multiplier becomes a divider.

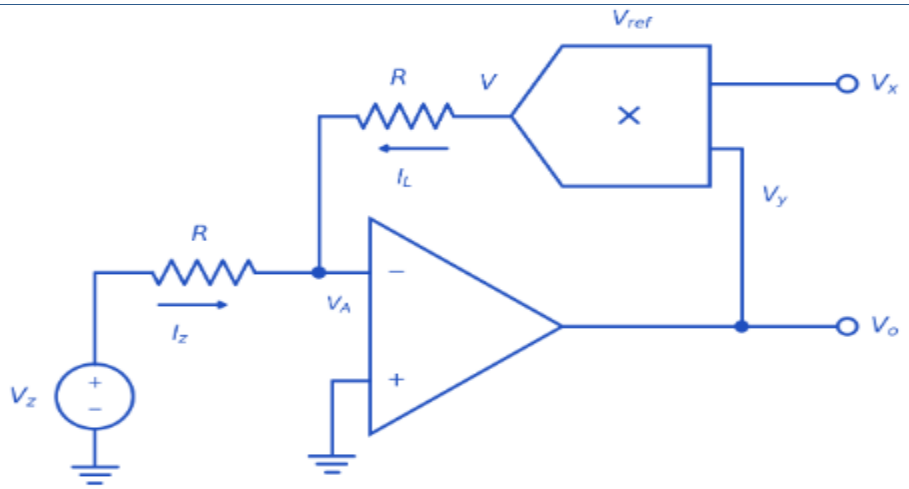


How it works

1. Log amps convert each input to its logarithm: $V_x \to K \ln V_x$, $V_y \to K \ln V_y$
2. Subtracting logs gives the log of the ratio: $\ln V_x - \ln V_y = \ln \frac{V_x}{V_y}$
3. The antilog amp undoes the log, so the output is the quotient: $V_o \propto \frac{V_x}{V_y}$

(Adder: $\ln V_x + \ln V_y = \ln(V_x V_y) \Rightarrow$ multiplier; Subtractor: $\ln V_x - \ln V_y = \ln(V_x/V_y) \Rightarrow$ divider)

Q. Find V_o for the circuit shown, in which a multiplier is placed in the feedback path.



Solution. From the multiplier, $V = \frac{V_x V_o}{V_{ref}}$, and $V_A = 0$.

KCL at node A:

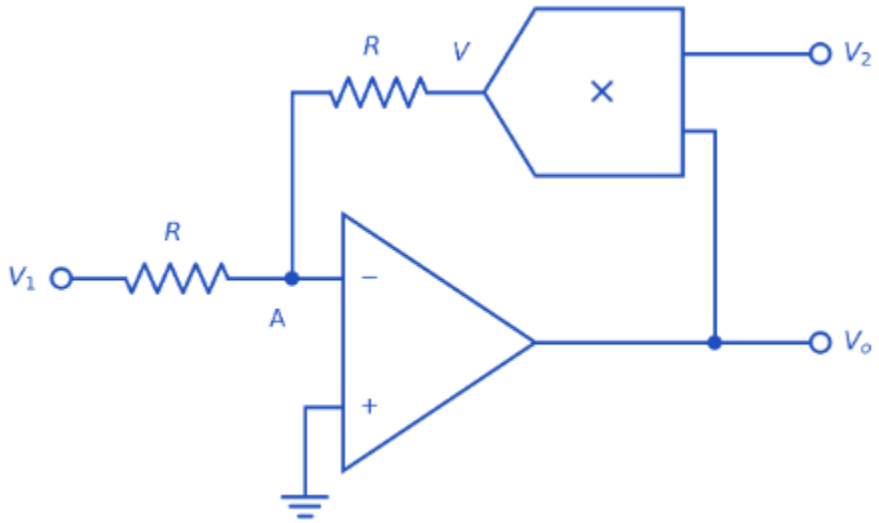
$$\frac{V_z}{R} + \frac{V}{R} = 0 \qquad \Rightarrow \qquad V = -V_z \qquad \Rightarrow \qquad \frac{V_x V_o}{V_{ref}} = -V_z$$

$$V_o = \frac{-V_z V_{ref}}{V_x} = -V_{ref} \times \frac{V_z}{V_x}$$

The above circuit is a **divider**.

Q7 — two-input multiplier

Q. The two-input multiplier shown has a scaling factor of 1 and produces a voltage output. If $V_1 = +15$ V and $V_2 = 3$ V, calculate the output voltage.



Solution.

$V = \frac{V_2 V_o}{V_{ref}}$ with $V_{ref} = 1$, so $V = V_2 V_o$.

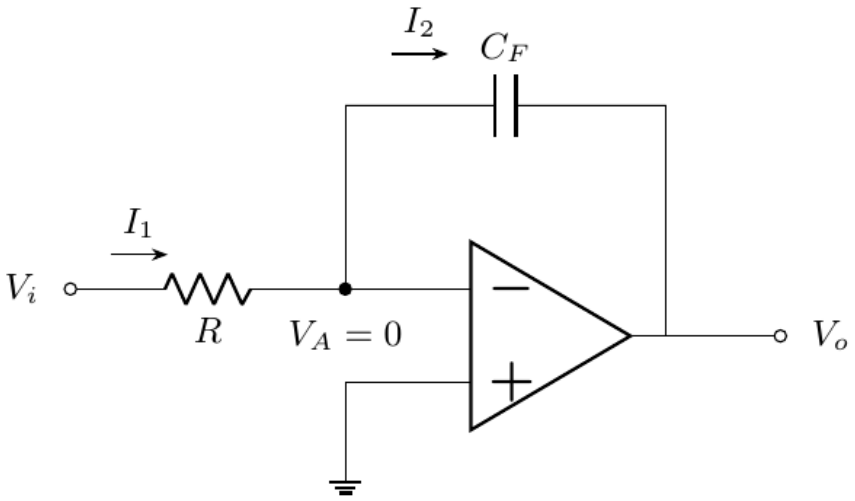
KCL at node A:

$$\frac{V_1}{R} + \frac{V}{R} = 0 \qquad \Rightarrow \qquad V = -V_1 \qquad \Rightarrow \qquad V_2 V_o = -V_1$$

$$V_o = \frac{-V_1}{V_2} = \frac{-15}{3} = -5 \text{ volts}$$

8. Integrator

A circuit in which the output voltage is proportional to the **time integration** of the input is called an integrator.



KCL at node A:

$$\frac{V_i - 0}{R} + C_F \frac{d}{dt}(V_o - 0) = 0 \qquad \Rightarrow \qquad \frac{dV_o}{dt} = \frac{-V_i}{RC_F}$$

$$V_o = \frac{-1}{RC_F} \int V_i \, dt$$

Input and output waveforms

V_i	V_o
square wave	triangular wave
spikes	square wave
sine	cosine

Gain of the integrator

$$\frac{V_o}{V_i} = A_{CL} = \frac{-Z_F}{Z} = \frac{-1/C_F s}{R} = \frac{-1}{RC_F s}$$

Putting $s = j\omega$:

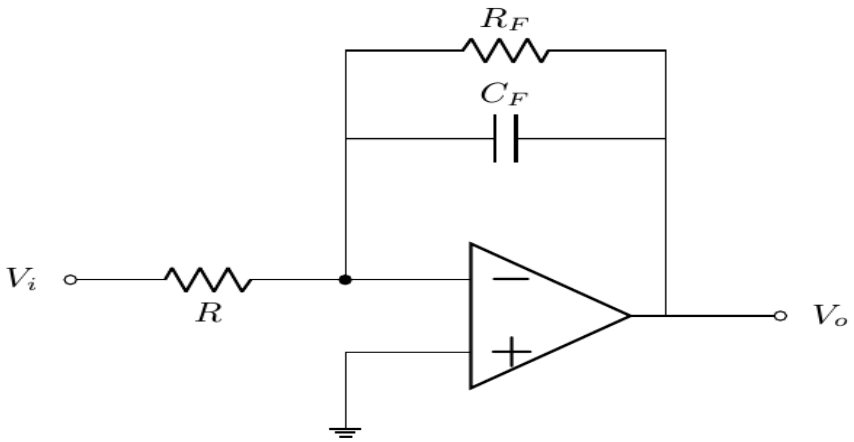
$$A_{CL} = \frac{-1}{j\omega RC_F} = \frac{j}{\omega} \times \frac{1}{RC_F} = \frac{j\omega_a}{\omega} = \frac{jf_a}{f}$$

$$|A_{CL}| = \frac{f_a}{f} \qquad \angle A_{CL} = 90° \qquad f_a = \frac{1}{2\pi RC_F}$$

f_a is the frequency at which the integrator provides **unity gain**.

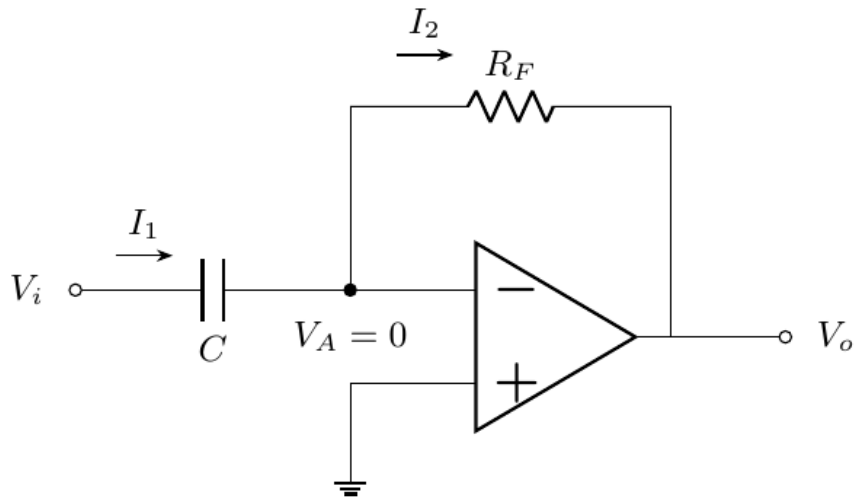
Practical integrator

If the input has a **non-zero DC component**, then the output of the integrator gets **saturated** and the circuit will not act as an integrator (a capacitor is an open circuit for DC). To prevent saturation of the output, a **resistor** should be connected in parallel with the capacitor C_F; such a circuit is called a **practical integrator**.



9. Differentiator

A differentiator provides an output voltage which is proportional to the **differentiation** of the input voltage.



KCL at node A:

$$I_1 + I_2 = 0 \qquad \Rightarrow \qquad C\frac{d}{dt}(V_i - 0) + \frac{V_o - 0}{R_F} = 0$$

$$V_o = -R_F C \frac{dV_i(t)}{dt}$$

Gain of the differentiator

$$\frac{V_o}{V_i} = A_{CL} = \frac{-Z_F}{Z} = \frac{-R_F}{1/Cs} = -R_F Cs$$

Putting $s = j\omega$:

$$A_{CL} = -R_F C \times j\omega = \frac{-j\omega}{\omega_a} = \frac{-jf}{f_a}$$

$$|A_{CL}| = \frac{f}{f_a} \qquad \angle A_{CL} = -90° \qquad f_a = \frac{1}{2\pi R_F C}$$

f_a is the frequency at which the differentiator provides **unity gain**. If $f \to \infty$ then $A_{CL} \to \infty$. Due to this infinite gain, a differentiator can be **unstable at high frequency**. A differentiator amplifies high frequency **noise**, which is undesired, as A_{CL} is directly proportional to f.

To prevent instability, a **resistor** can be connected in **series** with the capacitor; such a circuit is called a **practical differentiator**.

If the capacitor is replaced with an **inductor**, then an integrator becomes a differentiator and vice versa.